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Graph network learning for human skeleton modeling: a survey

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Abstract
Over the past few decades, Human Skeleton Modeling (HSM) has gained considerable attention in computer vision, exploring various practical applications such as the video surveillance, the human-computer interaction, the medical assistance analysis, and the autonomous driving through images and videos. The performance of HSM and its applications on challenging datasets has been significantly improved due to recent advancements of deep learning methods. These advancements have been extended to non-Euclidean or graph data with multiple nodes and edges. Because human joints and skeleton combinations are represented as graph structures, graph networks are appropriate for the non-Euclidean HSM. In recent years, graph networks have become essential tools for the HSM and behavioral analyses. However, prior surveys are often siloed, focusing either on a narrow class of models such as GCNs or on a single application like action recognition. A unified framework that systematically analyzes diverse graph network learning paradigms across the entire HSM pipeline has been notably absent. We conduct a survey of graph network methods for HSM and their application domains. This comprehensive overview includes a taxonomy of graph network techniques, a detailed study of benchmark datasets for HSM, extensive descriptions of the performance of graph networks in three major application domains, and a collection of related resources and open-source codes. Finally, we provided insightful recommendations for future research directions and trends of graph networks for HSM. This survey serves as the introductory material for beginners in graph network-based HSM and as the reference materials for advanced researchers.

Keywords Human skeleton modeling · Graph network · Pose estimation · Action recognition · Motion prediction

1 Introduction

Human skeleton modeling (HSM) serves as the foundation for numerous research efforts in the computer vision domain, with broad application prospects in action recognitions (Ahmad et al. 2021), human-computer interaction (Kosch et al. 2023), pose tracking (Ning

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et al. 2020), and motion prediction (Lyu et al. 2022). HSM aims to delineate the skeletal morphology within images, videos, and webcam streams, facilitating certain tasks such as action recognition, pose estimation, and motion prediction. Additionally, motion retargeting in computer graphics also relies on efficient and accurate skeletal topological relationship modeling, and HSM focuses on solving the long-distance dependence between skeletal points and non-Euclidean relationship problems in the design of graph networks. HSM provides technical support for character animation generation in computer graphics for the ability to capture the space-time characteristics of dynamic skeleton modeling and complex movements (Aberman et al. 2020; Mourot et al. 2023). This domain is a primary focus of scholarly investigation, both domestically and internationally, which can yield significant breakthroughs and numerous datasets, such as FLIC (Sapp and Taskar 2013), MPII (Andriluka et al. 2014), and MSCOCO (Lin et al. 2014). The frameworks for HSM include convolutional neural networks (CNNs) (Gamra and Akhloufi 2021), recurrent neural networks (RNNs) (Shi et al. 2019), graph convolutional networks (GCNs) (Yan et al. 2018), and graph neural networks (GNNs) (Li et al. 2021), substantially enhancing the HSM performance. HSM can be utilized to locate and identify the key joints of the human skeleton, connect them to form a projection on the two-dimensional plane of an image, and obtain the human skeleton structure. The primary task of the 3D HSM involves predicting the three-dimensional coordinates and angles of the human skeletal joints. Owing to the scarcity of annotated human skeleton datasets, most research methods have been based on existing HSM approaches.

Methods for HSM can be categorized into two main classes: traditional (Meyer et al. 2011) and deep learning-based methods (Lan et al. 2023). Before the widespread adoption of deep learning, datasets for the skeletal motion and high-performance computing platforms were not ubiquitous. Consequently, researchers have primarily employed the traditional computer vision and machine learning techniques for skeleton modeling. As shown in Fig. 1, the traditional methods typically rely on graph and pictorial structure models (Fischler and Elschlager 1973), which can consistently optimize the graph structure with constraints from human kinematics to estimate the human skeleton. Although the traditional methods exhibit the high temporal efficiency, they primarily rely on manually engineered features, such as the histogram of oriented gradient (HOG) (Dalal and Triggs 2005) and scale invariant feature transform (SIFT) (Al Ghamdi et al. 2012). This reliance on manually

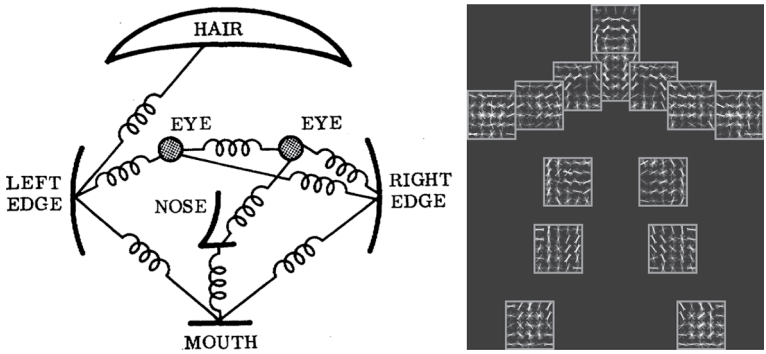

Fig. 1 Classical models for skeleton modeling. **a** Pictorial structures model (Fischler and Elschlager 1973). **b** Flexible mixture of components (Yang and Ramanan 2012)

engineered features fails to entirely leverage the image information, rendering the models sensitive to the variations in the appearance, viewpoint, occlusion, and inherent geometric ambiguity in the images.

The deep learning-based HSM leverages network models learned directly from image data to obtain the most effective representations. Deep neural networks (DNNs), primarily CNNs, can extract the convolutional features from images that are richer in semantic information, more accurate, and more robust than manually engineered features. By analyzing consecutive frames to capture the dynamics of human skeletal variations, deep learning methods establish more accurate topological structures between various key joints of the human skeleton and effectively handle subtle skeletal motion variations in complex scenarios. Deep learning has made significant advancements in HSM and its applications (Chen et al. 2023; Lyu et al. 2022; Gupta et al. 2022). Although the deep-learning-based skeleton modeling faces challenges such as insufficient data and occlusion, it has surpassed the traditional methods in terms of modeling accuracy and robustness (Chen et al. 2020).

To leverage the powerful representation capabilities of deep learning, conventional methods often rearrange skeleton data into the grid-like structures to implement traditional RNNs and CNNs directly. Although significant improvements have been observed in HSM, the capabilities of deep learning are limited. The state-of-the-art CNN-based HSM methods tend to miss detections, particularly in highly cluttered and fast-motion scenes. They inherently rely on initially detecting joints or humans using vision-based detectors, which can fail in challenging scenarios, such as crowded or blurry scenes, because of a lack of image evidence.

In recent years, graph networks have been applied to HSM (Yan et al. 2018; Wang et al. 2020; Mao et al. 2019), demonstrating a compelling advantage in addressing the HSM problems. The term "graph networks" in this paper represents both graph topologies and HSM modules. Skeleton data can convey close correlations between human joints, making graph networks more effective in capturing the dependencies between the joints, and more robust to complex scenarios such as viewpoint variations, occlusion, and self-occlusion. The human skeleton can naturally be modeled as a graph (Fig. 2), where the nodes represent the 2D or 3D positions of human joints and the edges represent skeletal connections. In some

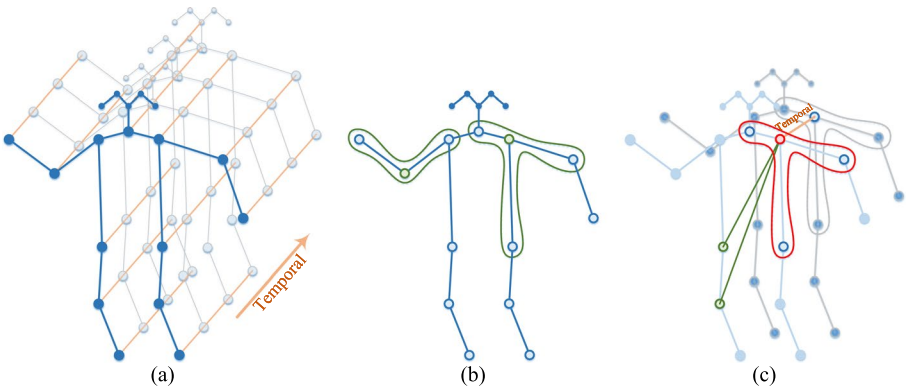

Fig. 2 Spatio-temporal graph for skeleton stacking. **a** Spatio-temporal graph of skeleton sequences. **b** Distance partitioning with root node distance. **c** Spatial partitioning: nodes are labeled based on their distance from the centroid of the skeleton. Red group represents the nodes directly connected to the red key point, and green line indicates the distance from the key points that is not directly connected to the red key point

cases, nodes are attributed with binary values (e.g., 0 or 1) to represent only the topology of the skeleton without explicit spatial coordinates. In graph networks, the adjacency matrix A determines the information passed between nodes, with the objective of obtaining features for each node by aggregating features from neighboring nodes during the learning process. For the target node, introducing the semantic information through the repeated transformation and aggregation of features from neighboring nodes can lead to a more robust feature representation and reduce the uncertainty in its position estimation. For instance, modeling a specific node in the graph, such as the right shoulder in Fig. 2c, may benefit from the information provided by both its immediate neighbors and nodes at multiple hop distances in the graph. The learning process of graph networks aims to acquire the contextual information for better HSM. Their performance, surpassing that of fully connected networks, demonstrates the crucial role of relational reasoning in alleviating deep ambiguities.

By utilizing the graph network learning methods, the structural and dynamic variations of human skeleton poses can be better captured and simulated to achieve more accurate and robust results in certain tasks such as pose estimation (Wang et al. 2020), action recognition (Yan et al. 2018), motion prediction (Mao et al. 2019), and pose tracking (Ning et al. 2020). This approach can improve the accuracy and robustness of the tasks related to human skeleton actions and the real-time performance and efficiency of systems. Constructing HSM and its applications based on graph network learning can facilitate the more accurate analysis and understanding of human poses and actions, providing strong support for applications in the human-computer interaction (Kosch et al. 2023), the medical image analysis (Fu et al. 2017; Bieth et al. 2017), and other related fields. Further research and development of HSM and applications based on graph network learning can propel the advancement of the computer vision field and provide superior solutions for tasks related to human skeletal actions.

To crystallize the evolutionary insights of these paradigms, consider a common challenge: estimating the pose of a person partially occluded by an object. A traditional method as shown in Fig. 1 relies on local appearance and would likely fail on the occluded parts due to the lack of visual evidence. A standard deep learning (CNN-based) method also struggles as it operates on a pixel grid and lacks an explicit model of biomechanical constraints to reason about the unseen. In contrast, a Graph Network-based method models the body as an interconnected structure as shown in Fig. 2. It overcomes this challenge by allowing visible joints to pass contextual information to occluded ones via message passing, performing relational reasoning to infer a plausible pose for the entire body. This fundamental shift from local-evidence processing to global-structural reasoning is the key advancement that motivates this survey.

To clearly position the unique contribution of this paper, we have systematically reviewed related surveys, as summarized in Table 1. A critical analysis of these works reveals a significant research gap. Existing surveys typically exhibit one or more limitations: they may focus on a single application domain, such as GCNs specifically for action recognition (Ahmad et al. 2021; Feng et al. 2022); they may cover a single task, such as various deep learning methods for pose estimation (Munea et al. 2020; Wang et al. 2021a); or, while mentioning graph networks, they often limit their scope to specific models like GCNs without covering the broader spectrum of graph network learning methodologies. To date, a comprehensive survey is missing that takes diverse graph network learning paradigms (beyond just GCNs) as its central theme and systematically examines their application across the entire

Table 1 Summary of surveys and reviews related to the human skeleton modeling and its applications in the past 5 years. DNN (deep neural network) broadly refers to deep learning-based methods

Title	Year	Venue	Framework	Theme	2D	3D
Deep learning-based action recognition with 3D skeleton: a survey (Xing and Zhu 2021)	2020	CAAI transactions on intelligence technology	DNN	Action recognition		√
The progress of human pose estimation: a survey and taxonomy of models applied in 2D human pose estimation (Munea et al. 2020)	2020	IEEE Access	DNN	Pose estimation	√	
A Comprehensive Study of Weight Sharing in Graph Networks for 3D Human Pose Estimation (Liu et al. 2020a)	2020	ECCV	GCN	Pose estimation		√
Monocular human pose estimation: a survey of deep learning-based methods (Chen et al. 2020)	2020	CVIU	DNN	Pose estimation	√	√
Monocular multi-person pose estimation: a survey (dos Reis et al. 2021)	2021	Pattern Recognition	DNN	Pose estimation	√	√
Bottom-up approaches for multi-person pose estimation and it's applications: a brief review (Kresović and Nguyen 2021)	2021	arXiv	DNN	Pose estimation	√	√
A review of deep learning techniques for 2D and 3D human pose estimation (Gamra and Akhloufi 2021)	2021	Image and Vision Computing	DNN	Pose estimation	√	√
Deep 3D human pose estimation: a review (Wang et al. 2021a)	2021	CVIU	DNN	Pose estimation		√
A comprehensive survey on 2D multi-person pose estimation methods (Wang et al. 2021b)	2021	Engineering applications of artificial intelligence	DNN	Pose estimation	√	
Graph convolutional neural network for human action recognition: a comprehensive survey (Ahmad et al. 2021)	2021	IEEE transactions on artficial intelligence	GCN	Action recognition	√	√
2D Human pose estimation: a survey (Chen et al. 2023)	2022	Multimedia systems	DNN	Pose estimation	√	
Deep Learning-Based Human Pose Estimation: A Survey (Zheng et al. 2023)	2022	ACM computing surveys	DNN	Pose estimation	√	√
A survey of top-down approaches for human pose estimation (Nguyen and Kresovic 2022)	2022	arXiv	DNN	Pose estimation	√	√
A review of 3D human body pose estimation and mesh recovery (Huang and Khan 2022)	2022	Digital signal processing	DNN	Pose estimation		√
3D human motion prediction: a survey (Lyu et al. 2022)	2022	Neurocomputing	DNN	Motion prediction		√
Human activity recognition in artificial intelligence framework: a narrative review (Gupta et al. 2022)	2022	Artificial intelligence review	DNN	Action recognition	√	√
A comparative review of graph convolutional networks for human skeleton-based action recognition (Feng et al. 2022)	2022	Artificial intelligence review	GCN	Action recognition	√	√

Table 1 (continued)

Title	Year	Venue	Framework	Theme	2D	3D
Skeleton graph-neural-network-based human action recognition: a survey (Feng and Meunier 2022)	2022	Sensors	GNN	Action recognition	√	√
Vision-based human pose estimation via deep learning: a survey (Lan et al. 2023)	2023	IEEE transactions on human–machine systems	DNN	Pose estimation	√	√
Human action recognition: a taxonomy-based survey, updates, and opportunities (Morshed et al. 2023)	2023	Sensors	DNN	Action recognition	√	√
Graph network learning for human skeleton modeling: a survey (OURS)	2024	–	Graph Networks	Human skeleton modeling	√	√

Human Skeleton Modeling (HSM) pipeline, from pose estimation to action recognition and motion prediction.

This survey is designed to fill this gap. By providing a unified and principled perspective, we connect the foundational theory of graph networks to multiple cutting-edge applications in HSM. Compared with the existing literature, the main contributions of this study are summarized as follows:

1. This survey provides a broad and detailed summary of graph network-based HSM and its applications, encompassing the recent advancements in skeleton modeling, action recognition, and pose estimation. This overcomes the limitation of previous surveys that focused on a single application.
2. We propose a novel taxonomy for graph networks in HSM that is based on the "locus of innovation." This framework categorizes methods into "graph structure-based frameworks" and "HSM-module-based frameworks." This approach moves beyond previous simple, application-based or model-specific groupings to provide a more fundamental and systematic framework for understanding the evolution and intrinsic connections of different technical routes in the field.
3. We covered the recent and advanced research on graph network learning methods for visual HSM and their applications, providing the detailed introductions and insights into relevant studies. This resource could offer readers access to the state-of-the-art models.
4. We provided the comprehensive evaluation and comparison of existing methods using various datasets as well as the case analyses for different application scenarios. Through these evaluations, readers can gain a more thorough understanding of the performance and effectiveness of the visual HSM methods and graph network learning applications in diverse scenarios.

To ensure a systematic and comprehensive overview, this survey was compiled following a structured review methodology. The scope of this work is centered on graph network learning methods applied to vision-based human skeleton modeling. We explicitly exclude methods based on non-visual sensors and those that do not utilize a graph network as a core component. Our temporal focus is on the modern deep learning era, primarily covering influential papers published from 2018 (marked by the advent of ST-GCN) to 2025.

Our literature selection criteria involved a multi-stage process. First, we conducted an extensive search on major academic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, and arXiv, using a combination of keywords such as "graph neural network", "graph convolutional network", "human skeleton", "pose estimation", "action recognition", and "motion prediction". The initial pool of papers was then filtered based on title and abstract relevance. Finally, we conducted a full-text review of the shortlisted papers and employed a technique, where we recursively traced the citations of key papers to identify further relevant works.

Evaluation approach for the selected literature was inductive and taxonomy-driven. Rather than a simple chronological or application-based summary, we performed a thematic analysis to identify the fundamental locus of innovation within each paper. This iterative process of comparing and grouping core contributions led to the development of our proposed taxonomy, which distinguishes between innovations in graph structure and innovations in operational modules. This approach provides a more insightful framework for understanding the field's evolution.

The structure of this study is illustrated in Fig. 3. Section 3 discusses the five representations of the graph networks employed in HSM methods. Section 4 presents an overview of the datasets for HSM and their applications along with evaluation metrics. Section 5 elucidates the current applications of the graph-network-based human skeletal models for pose estimation, action recognition, and motion prediction. Finally, Sect. 6 provides a critical discussion of the factors affecting the application of graph networks in skeletal modeling and explores the potential future trends.

2 Introduction to graph networks for HSM

Graph networks have emerged as a powerful tool for processing graph-structured data, making them particularly suitable for HSM. Unlike traditional neural networks designed for grid-like data such as images or sequences, graph networks leverage the relational infor-

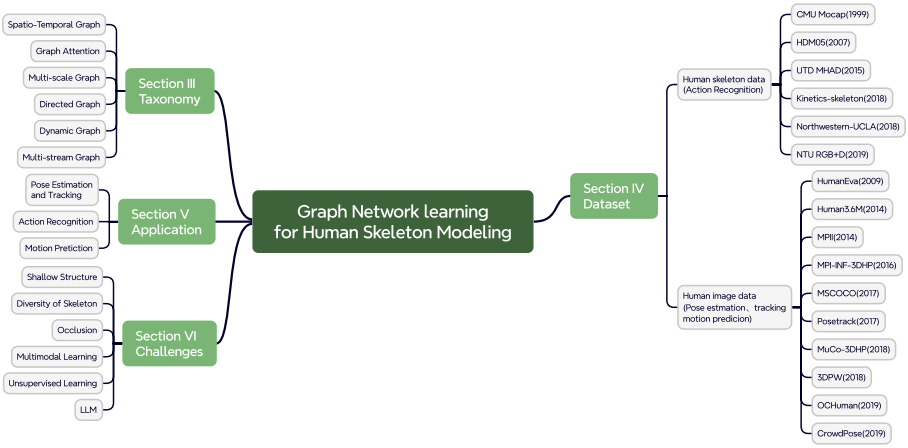

Fig. 3 Framework of this study, which introduces the taxonomy of graph network learning, various datasets, and applications for human skeleton modeling

mation in graphs. In the context of HSM, the human skeleton is naturally represented as a graph, where joints correspond to nodes and physical or functional connections between joints are represented as edges. This representation allows Graph Networks to capture the intricate spatial and temporal dependencies inherent in skeletal motion data.

At the core of graph networks is the message passing paradigm, which enables each node (i.e., a joint) to iteratively update its feature representation by aggregating information from its neighbors. This process precisely mimics the inter-dependencies among joints in human motion–for instance, the state of the wrist is clearly influenced by the states of the elbow and hand. This process can be described by a general graph convolution equation:

$$h_i' = \sigma(\sum_{j \in \mathcal{N}(i)} A_{ij} W h_j + b), \tag{1}$$

where, h_j is the feature vector of a neighboring joint v_j. In HSM, this can be a multi-dimensional vector containing its 3D coordinates, velocity, or orientation, describing the joint's current kinematic state. $\mathcal{N}(i)$ represents the set of neighboring nodes for joint v_i, typically defined by the inherent physical connections of the human body. A_{ij} is an element from the adjacency matrix, defining the "weight" or "strength" of the information passed from joint v_j to v_i. This can be a simple binary value (indicating physical connection) or a learnable parameter that expresses the importance of different connections (e.g., the shoulder-elbow connection is more structurally significant than a non-physical wrist-head connection). W is a learnable weight matrix that acts as a "feature transformer," applying a linear transformation to the neighbor's features h_j before they are aggregated. Functionally, it learns how to extract the most relevant patterns from the raw states of the neighbors for the target task. The $\sum_{j \in \mathcal{N}(i)}$ is the aggregation step, which gathers all the transformed information from the neighbors. This simulates the process where the target joint considers the states of all its relevant body parts. b is a bias term, and σ is a non-linear activation function. As human motion is a highly complex, non-linear process, the activation function is crucial for capturing this complexity. h_i' is the updated feature vector for joint v_i. After one round of message passing, it now incorporates contextual information from its local neighborhood, leading to a more robust and richer representation of its own state.

To extend this capability from a single-frame spatial structure to a continuous video sequence, spatio-temporal graph convolutional networks (ST-GCNs) were proposed. They capture the dynamic evolution of the skeleton by processing spatial and temporal dimensions in parallel. A single layer's operation can be distilled as:

$$H' = \sigma\left(AHW_s\right) + \mathrm{Conv1D}(H, W_t), \tag{2}$$

where, $H \in \mathbb{R}^{N \times T \times C}$ is the input tensor for the entire skeleton sequence, representing the C-dimensional features for N joints across T time frames. The two key components of Eq. 2 address:

Spatial graph convolution $\sigma(AHW_s)$ operates independently within each time frame, focusing on capturing the intra-frame spatial configuration as shown in Fig. 2b. The adjacency matrix A defines the physical joint connectivity, and W_s is the learnable weight for the spatial dimension. It learns the relative positioning of body parts by aggregating information from all joints.

Temporal convolution $\mathrm{Conv1D}(H, W_t)$, on the other hand, applies a standard 1D convolution along the time axis for each joint's feature sequence. It focuses on capturing the inter-frame temporal dynamics as shown in Fig. 2a. W_t is the temporal convolution kernel. It learns the motion patterns (like velocity and acceleration) for each joint independently.

Finally, by fusing the spatial configuration features and the temporal dynamic features via addition, the model obtains a comprehensive spatio-temporal feature representation H'. By stacking multiple such ST-GCN layers, the network learns complex human motion patterns ranging from low-level to high-level and from local to global, providing a powerful feature foundation for subsequent tasks like action recognition, pose estimation, or motion prediction. This visual representation highlights how graph networks operate at the intersection of spatial and temporal domains to model human skeleton data effectively.

While graph networks provide a powerful theoretical framework for HSM, their basic forms also reveal inherent limitations when facing real-world complexities. The effectiveness of these models relies heavily on the completeness and accuracy of the input skeleton graph data. In practical scenarios, challenges such as occlusion are extremely common. When body joints are occluded, their corresponding node features can be missing or noisy, which severely disrupts the message-passing mechanism that graph networks depend on and degrades the accuracy of the overall pose. Similarly, significant viewpoint variations pose another challenge. A Graph network trained on a specific view may struggle to generalize to a drastically different one because the projected spatial relationships and distances between joints change fundamentally, potentially confusing a model with fixed relational assumptions. Furthermore, a basic graph network typically employs a predefined and static graph topology, which constrains its expressive power. For actions like hugging or shaking hands, a strong, temporary functional relationship is established between the hands, yet no such edge exists in the standard skeleton topology. These limitations highlight the need for more advanced and adaptive graph network architectures to improve robustness and expressiveness in complex, real-world environments. The drive to overcome these very challenges has motivated the development of the innovative frameworks that we will systematically categorize and discuss in the following section.

3 Graph networks taxonomy for HSM

The application of graph networks in HSM primarily involves encoding skeletal information and embedding it into latent representations of human behavior. Graph networks aim to capture the contextual dependencies and intrinsic topological characteristics of human behavior while incorporating the spatio-temporal information for modeling. In the graph-network-based HSM, the first step is to construct the graph structure of the human skeleton by connecting joints to form the nodes and edges of the graph, where the nodes represent the skeletal joints (head, arms, legs, etc.), and the edges represent the connections between them. The graph networks can be utilized to extract features and learn from this graph. By applying convolution operations to each node, graph networks can adopt the connections between the nodes to propagate and aggregate information, capturing the structure and dynamic variations of the human skeleton (Li et al. 2018). Through iterative computations, these methods can extract the high-level semantic information from input features (Zhao et al. 2019). The graph network learning methods effectively model and learn the structure

and dynamics of the human skeleton, capture the spatio-temporal relationships, and offer the flexibility and scalability for analyzing and understanding human poses and actions. This approach also opens up new possibilities for applications such as human-computer interaction (Kosch et al. 2023) and virtual reality (Liu et al. 2018).

Before detailing the taxonomy, it is necessary to first elaborate on our categorization criteria. This criterion is the identification of the primary locus of innovation within each research work. It distinguishes between two fundamental levels of contribution: innovations in the definition of the graph itself, and innovations in the design of modules that operate on the graph. As the field matures, it is increasingly common for advanced methods to be hybrid models that exhibit characteristics from multiple subcategories (Si et al. 2019; Shi et al. 2019; Huang et al. 2023; Zhu et al. 2023; Liu et al. 2020b; Zhang et al. 2023; Shi et al. 2020; Li et al. 2022, 2021; Chen et al. 2021; Li et al. 2021, 2020; Gao et al. 2023). To handle such cases and ensure a consistent classification, our classification prioritizes the authors' stated contribution, the most architecturally novel component, and evidence from ablation studies to ensure a systematic and consistent framework. In such cases, our classification prioritizes the innovation that is most central to the paper's contribution as highlighted by its authors or as judged by its impact on the field. This systematic approach aims to provide readers with a clear and logically consistent perspective on the evolution of this research area.

Following these criteria, we propose a taxonomy for graph network methods in HSM. We categorize the existing methods into two primary frameworks, as illustrated in Fig. 4.

Graph structure-based frameworks The core contribution of these methods is to redefine or dynamically evolve the graph topology itself. They move beyond using a predefined, static graph based on physical connections and explore how to construct a more expressive graph. The innovation centers on the design of the adjacency matrix A.

1. *Directed graph networks* Model causal or hierarchical dependencies between joints by introducing an asymmetric adjacency matrix, where $A \neq A^T$. DGNN (Shi et al. 2019) specifically designed a directed GNN to extract information about joints, bones, and their relationships.
2. *Dynamic graph networks* Focus on capturing the changing relationships between joints during motion by making the adjacency matrix time-variant, denoted as A_t. Skeletal-GNN (Zeng et al. 2021) proposes a time-aware dynamic graph construction method,

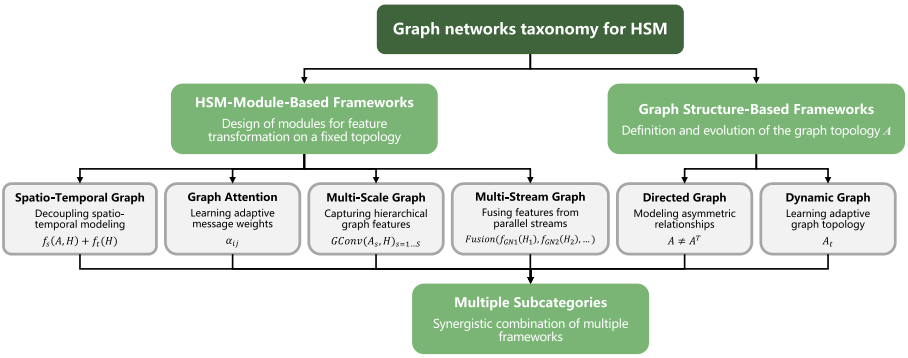

Fig. 4 Graph networks taxonomy diagram for HSM

while LDRGCN (Cui et al. 2020) uses a dynamic graph to implicitly learn links between geometrically separated joints.*HSM-module-based frameworks* These methods typically operate on a relatively fixed graph topology. Their core contribution lies in designing novel neural network modules to more effectively extract, aggregate, and enhance node features. The innovation centers on the design of the feature transformation function.

1. *Spatio-temporal graph networks* Design specific modules to separately handle the spatial dimension (intra-frame joint relations) and the temporal dimension (inter-frame joint dynamics), followed by an effective fusion. The pioneering ST-GCN (Yan et al. 2018) first applied GCNs to action recognition and established this paradigm of separate spatio-temporal processing.
2. *Graph attention networks* Design attention modules to assign adaptive weights to messages from different neighbors by learning attention coefficients α_{ij}. GAST-NET (Liu et al. 2020b) utilizes graph attention mechanisms to effectively learn skeletal structure, local kinematic connections, and symmetry.
3. *Multi-scale graph networks* Design modules that can simultaneously learn features at different receptive fields (joints, limbs, and the whole body). MST-GCN (Chen et al. 2021) enriches the model's receptive field via multi-scale spatio-temporal graph convolutions, while MS-G3D (Liu et al. 2020c) directly models dependencies across spatio-temporal nodes at multiple scales.
4. *Multi-stream graph networks* Design parallel network streams to process different modalities of input before fusing the information. 2 s-AGCN (Shi et al. 2019) employs a dual-stream framework to model both first-order and second-order skeleton data, while MS-AAGCN (Shi et al. 2020) models information about joints, skeletons, and their actions simultaneously within a multi-stream framework.

Before presenting this taxonomy in detail, it is essential to clarify its advantages over existing organizational frameworks for these methods. Previous approaches to categorizing the literature have often implicitly relied on two main schemes: one is application-based, dividing methods into categories like action recognition or pose estimation; the other is model-family-based, grouping together variants of a specific architecture such as ST-GCN. However, these approaches have limitations. An application-based grouping obscures underlying methodological differences, while a model-family-based grouping can devolve into a "laundry list" of methods that lacks high-level insight.

The "locus of innovation" taxonomy proposed in this study offers a more fundamental and insightful perspective. Its superiority is threefold: first, it moves beyond a surface-level focus on a model's application to an essential analysis of its core innovative mechanism, distinguishing between innovations in graph structure versus those in operational modules. Second, this systematic framework helps reveal deeper developmental trajectories in the field, showing an evolution from an initial focus on module-based innovations on fixed graphs toward more recent and fundamental work on the graph structure itself. Therefore, this taxonomy serves not merely as an organizational tool, but as an analytical one, enabling readers to better understand the logical evolution of the field and its future directions.

3.1 Spatio-temporal graph networks

Spatio-temporal graph networks fall under the category of "HSM-module-based frameworks." The core design principle of this category is to decouple the spatial and temporal dimensions of the graph, process them separately, and then effectively fuse the resulting features. As shown in Fig. 5, these models utilize the natural topological graph structure of human skeletons to capture the spatial and temporal dependencies, enhancing their modeling capability for human poses and actions. Spatio-temporal graph networks aggregate skeletal joints across different time steps by performing CNN operations in the temporal dimension, utilizing the temporal relationships for the feature extraction and learning. This approach can effectively encode the dynamic variations in human skeletons and capture the joint positions and the motion patterns across different frames.

One prominent model in this field is the spatio-temporal GCN (ST-GCN) (Yan et al. 2018). The authors first applied the GCN in the action recognition and proposed an ST-GCN for the general modeling of dynamic skeletons. The ST-GCN treated human skeletal joints as the nodes, spatially connected the joints in the human skeleton as the spatial edges, and temporally aggregated the corresponding nodes across previo us and subsequent frames as the temporal edges. A spatio-temporal graph is represented as, $\mathrm{G} = \{G_1, G_2, \cdots, G_N\}$, where $G_1, G_2, \cdots, G_N$ represent the input skeleton graph in each frame, and N denotes the total number of frames. Subsequently, multiple layers of the spatio-temporal GCN are constructed on this spatio-temporal graph to extract information separately in the spatial and temporal dimensions. This model has been widely applied to action recognition tasks to achieve the state-of-the-art performance on large datasets. Furthermore, several improvements to ST-GCN have been proposed to enhance the recognition accuracy, including MST-GCN (Chen et al. 2021), MST-GNN (Li et al. 2021), ST-BLN (Heidari and Iosifidis 2021), WST-GCN (Miki et al. 2020), Stacked-STGCN (Ghosh et al. 2020), DSTD-GC (Fu et al. 2022), ST-PGN (Parsa et al. 2020).

SIhGCN (Setiawan et al. 2022) feeds the spatio-temporal graph representation into a graph convolutional layer (GCL) to extract the graph structure into spectral feature representations. MMNet (Yu et al. 2023) utilizes the attention weights learned by ST-GCN to

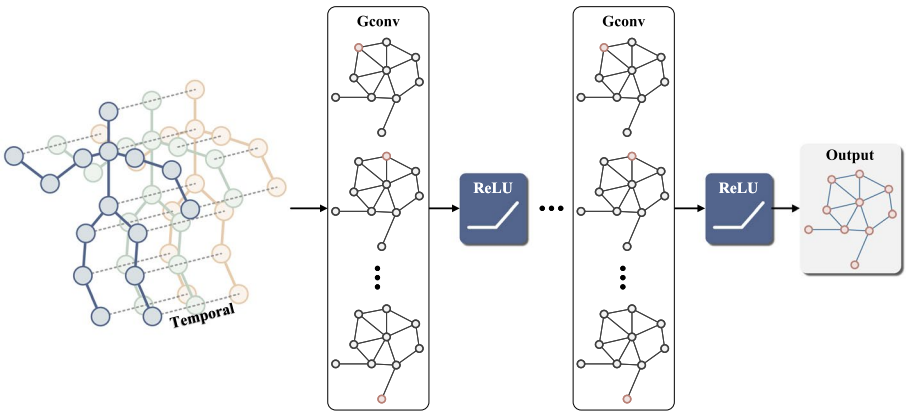

Fig. 5 Framework of spatio-temporal graph networks. The human skeleton is considered as a fixed graph with up to N frames. The spatial relationships between joints are established using intra-frame spatial edges, whereas the temporal relationships between joints are established using inter-frame temporal edges

focus on the spatio-temporal regions of interest (STROI) in the RGB modality. SDGCN (Wu et al. 2019) introduces the spatial residual layers and the dense connection blocks to capture rich spatio-temporal information. Prior applications of GCNs in modeling the temporal dynamics and the spatial articulation interactions have caused the issue of cross-contamination between motion and spatial correlations. To address this, STSGCN (Sofianos et al. 2021) is introduced as the first spatio-temporal separable GCN, which considers the spatio-temporal connectivity in the affinity matrix and inhibits the spatiotemporal contamination while achieving the entire joint-to-joint and spatio-temporal correlations. The aforementioned research demonstrates that ST-GCN can efficiently utilize the temporal structure of skeleton data to accurately parse complex motion patterns. Its advantage lies in the effective fusion and modeling of spatio-temporal information to capture complex spatio-temporal relationships in action sequences through GCNs. ST-GCN can handle the action sequences of different lengths and performs well in processing the variable-length sequences, providing a flexible solution for dynamic scenarios in practical applications.

Some researchers have combined long short-term memory (LSTM) (Shu et al. 2021) and graph networks within a spatio-temporal framework to improve the modeling of human skeletal structures. SR-TSL (Si et al. 2018) integrates a spatial reasoning network (SRN) and a temporal stack learning network (TSLN), where SRN captures the advanced spatial structural information within each frame through residual GNNs and TSLN models the detailed temporal dynamics of the skeletal sequence using skip-clipped LSTMs. AGC-LSTM (Si et al. 2019) captures the discriminative features in both the spatial configuration and temporal dynamics by exploring the co-occurrence relationships between the spatial and temporal domains. BGC-LSTM (Zhao et al. 2019) extracts the structure-aware feature representations from the pose data using GCN and captures the temporal dynamics through LSTMs. LSGM (Huang et al. 2020) introduces a graph temporal spatial calibration (GTSC) module that combines LSTM and GCN to learn the advanced spatio-temporal features from skeletal data for the improved discriminative power. RHCN+ACSC+STUFE (Li et al. 2020) is a multi-level co-occurrence graph convolution LSTM (MCGCLSTM) model that integrates GCN into LSTM to enhance the utilization of skeletal structure information and multi-level co-occurrence feature learning. CCG-LSTM (Tang et al. 2022) explores the motion characteristics of individuals in group activities by proposing the spatio-temporal consistency constraints (STCC) and global consistency constraints (GCC) to capture and quantify the related motion.

The advantage of GCN+LSTM lies primarily in its comprehensive modeling of spatio-temporal relationships. LSTM effectively captures the evolution of skeletal motions over time and handles the long-term temporal dependencies to describe the relationships between skeletons across video frames. Meanwhile, GCN accurately models the spatial relationships between the skeletal nodes, providing the in-depth understanding of node interactions. The combination of LSTM and GCN integrates both the temporal characteristics of the action sequences and spatial correlations between the skeletal nodes, improving the accuracy of human skeletal action modeling.

Different from both methods mentioned above, researchers have explored the spatio-temporal modeling of human skeletal structures. MS-TA-HGCN-FC (Huang et al. 2023) are the spatial-temporal stream GCNs where the spatial stream learns the spatial relationships and the temporal stream learns the temporal dynamics, utilizing the adaptive GCNs to focus on the nodes of different ranges across layers and combining the spatio-temporal refinement

through a dual-branch structure. DGNN (Shi et al. 2019) uses GNNs and RNNs to learn the temporal bone topology, addressing the variations in skeletal structure across different time frames. MS-G3D (Liu et al. 2020c) proposes a unified spatio-temporal GCN that employs the dense cross-spatio-temporal edges as the skip connections to directly propagate information on the spatio-temporal graph. MLSTGNet (Zhu et al. 2023) decomposes the learning of human skeletal structures into a general graph and individual graphs, utilizing the spatial data-driven excitation (SDE) module for different joint relationships and the temporal motion excitation (TME) module for motion-sensitive features. PGBIG (Ma et al. 2022) introduces a network composed of spatial dense GCN (SDGCN) and temporal dense GCN (TDGCN) to extract the spatio-temporal features across the global receptive field of the entire pose sequence. STF-Net (Wu et al. 2023) explicitly models the multi-granular contextual information between skeletal joints and parts to capture local sensitive joints of temporal dynamics. These methods demonstrate effective performance in modeling complex human dynamics by learning spatial structures and temporal variations using the spatio-temporal GCNs. The selection of the appropriate architecture depends on the specific application requirements and task objectives.

3.2 Graph attention networks

Graph attention networks (GATs) are categorized as "HSM-module-based frameworks", with a design focus on employing adaptive attention mechanisms to dynamically adjust the weights of nodes and edges. These networks emphasize modeling critical skeletal nodes and significant connections. Figure 6 illustrates that they can enhance the focus on the key nodes and skeletal connections, strengthening the representation of these key features during the feature extraction process and capturing their dependencies. The introduction of graph attention mechanisms allows the model to prioritize the important skeletal joints and connections for improving the accuracy and robustness of the feature extraction. GATs can dynamically calculate the attention scores between the nodes by learning the relationship weights and applying these scores to the weighted aggregation of features.

The adaptive learning of skeletal structures can be achieved through attention mechanisms, which automatically learn the spatial and structural relationships between skeletal joints and emphasize the key joints for accurate pose estimation through weight allocation,

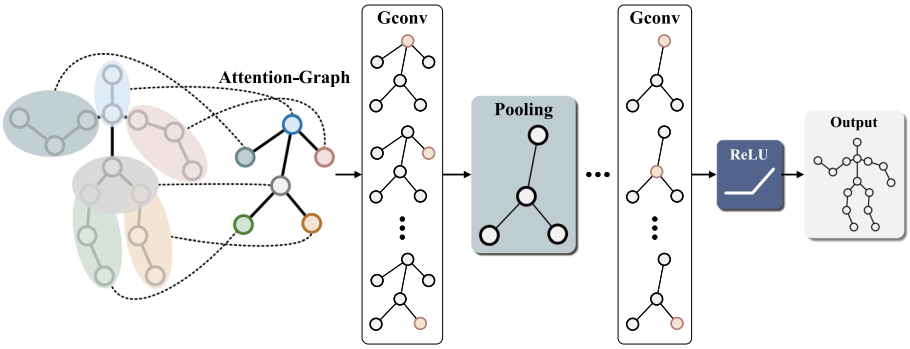

Fig. 6 Framework of graph attention networks (GATs). GATs incorporate both graph and attention networks into their architectures. Attention networks are used to emphasize the significant parts of the body skeleton and are mostly implemented using LSTM or attention modules

eliminating the need for manually designing the structural priors. For instance, the position information of major skeletal joints is often more crucial because of their distinct motion patterns. The attention mechanism enhances the robustness and accuracy of HSM by learning these structural relationships and weight assignments. GAST-NET (Liu et al. 2020b) utilizes the graph attention mechanisms and convolution operations to integrate the spatial and temporal information, consisting of the interleaved temporal convolutional blocks and the graph attention blocks that collectively improve the learning of skeletal structures, local kinematic connections, symmetry, and capture both temporal dependencies and spatial semantics effectively. AGC-LSTM (Si et al. 2019) employs attention mechanisms to enhance the key joint information in each AGC-LSTM layer, focusing on important joints that contribute to human skeletal motion and enhancing their spatial information. STGAT (Hu et al. 2022) uses graph attention mechanisms to adaptively compute the edge weights in local spatio-temporal graphs for aggregating more representative features. EfficientGCN (Song et al. 2023) proposes a spatial-temporal joint attention mechanism (ST-JointAtt) based on GCN to focus on important joints and frames. In LKA-GCN (Liu et al. 2023), attention mechanisms are crucial for capturing the long-range dependency relationships between human joints, balancing the importance of time factors to focus on information-rich temporal features, and capturing the advanced spatio-temporal patterns. Moreover, the attention mechanisms can generate the attention maps that reflect the importance of each joint position to identify actions more accurately.

Diverse relationship modeling is crucial in HSM, encompassing not only the spatial dependencies between skeletal joints, but also certain complex relationships such as angles, length ratios, and high-level semantics. Different types of dependencies reflect various aspects of the pose structure, all of which should be modeled. GATs provide a multi-head mechanism, allowing each attention head to model a specific type of dependency. KA-AGTN (Liu et al. 2022) enhances the high-order spatial dependencies between joints through the GCN-based attention mechanism that extracts the spatio-temporal features of skeletons. AAM-GCN (Xie et al. 2021) introduces dimension attention mechanisms to strengthen important information in different dimensions of skeleton data while resolving the oversmoothing problem in GCN with attention adjacency matrices, eliminating the dependence on the manually designed centroids. SCAFF (Li et al. 2022) proposes a semantic correlated attention module (SCAM) to learn the semantic correlations between skeletal parts and motion time for feature fine-tuning. PHGANet (Zhang et al. 2023) employs the Hop-Aware skeletal graph attention (HSGAT) module across multiple parallel branches of stacked modules to learn the rich jump-aware correlations of human skeletal structures at different scales. MS-AAGCN (Shi et al. 2020) introduces a multi-stream attention-enhanced adaptive GCN (MSAAGCN) that utilizes the spatio-temporal channel (STC) attention mechanisms to enhance the adaptive graph convolution layers, focusing more on the second-order information of important joints and frames.

3.3 Multi-scale graph networks

Multi-scale graph networks are part of the "HSM-module-based frameworks", which aim to capture both fine-grained local features and global dynamic structures by designing multi-scale graph representations that integrate local and global skeletal relationships. As illustrated in Fig. 7, these networks effectively capture fine-grained information related to

human skeleton by introducing multi-scale feature representations and convolutional operations on a graph structure. By modeling the relationships between skeletal joints at different scales, multi-scale graph networks can offer a more comprehensive description of the spatio-temporal characteristics of human actions. A particular multi-scale graph network first constructs a graph model that encodes the skeletal structure information, with the nodes representing the key skeletal joints and the edges representing the connections between them. The network then defines neighborhoods at various spatial and temporal scales to learn the node features within each scale. For instance, graph networks can operate within 1-hop or 2-hop neighborhoods of skeleton nodes. The larger-scale convolutions capture the global structures, whereas the small-scale convolutions learn local details. Finally, the network integrates feature representations from different scales through operations. By fusing this multi-scale information, the model can obtain a more comprehensive understanding of the skeletal structure.

Multi-scale spatio-temporal graphs have been utilized in various models to capture the features of human skeletons at different spatial and temporal scales. In MS-AAGCN (Shi et al. 2020), different receptive fields are employed to capture the features of human skeleton at varying spatial scales. MSR-GCN (Dang et al. 2021) aggregates and decodes the features extracted at each scale to obtain the residuals between the input and target skeleton, further exploring the multi-scale characteristics of human skeletons. MST-GNN (Li et al. 2021) introduces multi-scale spatial graphs and multi-scale temporal graphs, represented by multiple subgraphs of different spatial and temporal scales, where the spatial subgraph represents the relationships between the human skeletal nodes and the temporal subgraph represents the temporal continuity relationships. MST-GCN (Chen et al. 2021) consists of multi-scale spatial graph convolution (MS-GC) and multi-scale temporal graph convolu-

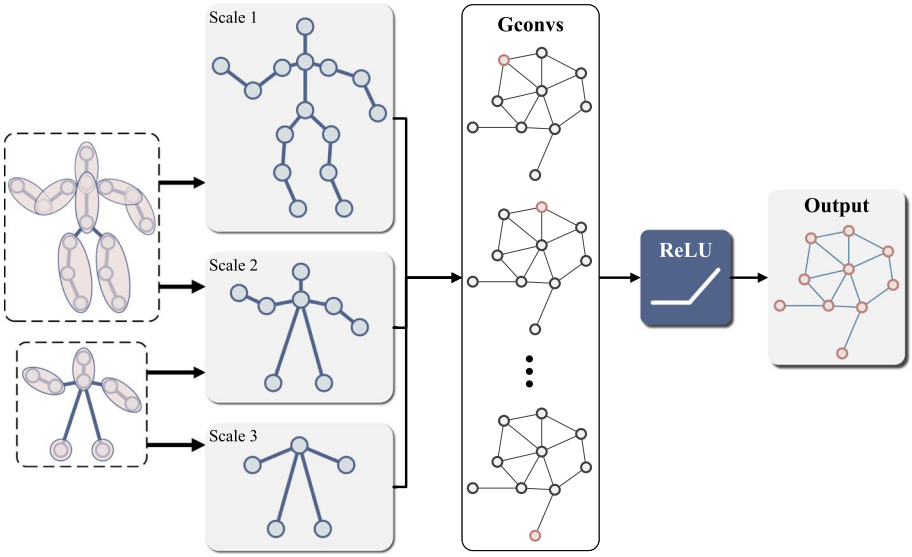

Fig. 7 Framework of multi-scale graph networks. By grouping nearby joints and replacing the group with a single joint, the human pose can be gradually abstracted to obtain a series of multi-scale poses ranging from fine to coarse. The initial GCN maps the multi-scale pose inputs from the pose space to the feature space, the intermediate GCN is used to extract features in the feature space, and the final GCN maps the features back to the pose

tion (MT-GC) modules, which stack multiple blocks to cooperate in learning the effective representations of human skeletal motion, decomposing the corresponding local graph convolutions into a set of subgraph convolutions to enrich the receptive field of the model in both spatial and temporal dimensions. MS-G3D (Liu et al. 2020c) directly models the dependencies across the spatio-temporal nodes, which can facilitate the unhindered spatio-temporal information flow and capture complex spatio-temporal dependencies. MLSTGNet (Zhu et al. 2023) decomposes the learning of human skeletons into a general graph and a single graph, whereas the simplified multi-scale temporal convolution (MSTCN) network enriches the expressive power of temporal features. SGSN (Li et al. 2021) proposes a dynamic multi-scale GNN (DMGNN) for predicting human motion based on 3D skeletons, which is adaptive during the training and dynamic between network layers. Additionally, the local–global GCN (LGGCN) proposed by Graph2Net (Wu et al. 2022) introduces the multi-faceted spatial-aware representations, and the region-aware GCN (RAGCN) embeds the regional relationships into the spatio-temporal graph models.

SCAFF (Li et al. 2022) introduces multi-scale feature fusion modules (MSFFs) designed to capture the motion characteristics of observed human states. DMGNN (Li et al. 2020) constructed a comprehensive paradigm for modeling human skeletal joint relationships by encompassing the local associations between skeletal joints at the same scale and the remote dependencies between skeletal structures at different scales. The multi-scale graph computation units (MGCU) in the encoder extract features from multi-scale graphs to capture the semantic information from multiple anatomical levels, generating skeletal motion representations. Motif-GCN (Wen et al. 2019) utilizes GCN to encode the hierarchical spatial structures, employing the variable temporal density blocks to leverage the local temporal information across different scales in human skeleton sequences. The improved SAMotif-GCN (Wen et al. 2023) uses GCNs to capture both the local and non-local relationships in the spatial and temporal domains, exploiting the latent relationships between joints by non-physically connecting them to impose the high-order locality and assigning different semantic roles to the physical neighbors of joints for the hierarchical structure encoding. HetGCN (Gao et al. 2023) dynamically analyzes the pairwise interactions and the heterogeneous context aggregation between each node and its spatio-temporal neighbors to capture rich semantic representations from skeleton-based graphs.

3.4 Multi-stream graph networks

Multi-stream graph networks belong to the “HSM-module-based frameworks” category, and are specifically designed to handle methods that integrate skeleton graph information with other data sources or modalities. The core architectural idea is to process multiple, different sources of input information in parallel and then fuse them to obtain a more robust feature representation. As shown in Fig. 8, these “streams” can process different orders of information derived from the same skeleton, such as one stream for joint coordinates and another for bone vectors representing length and orientation, or they can process data from different modalities entirely. Each stream extracts features independently, and these features are combined at a later stage of the network via concatenation, addition, or other fusion strategies. This approach is based on the principle of information complementarity and aims to boost overall model performance by leveraging diverse input perspectives.

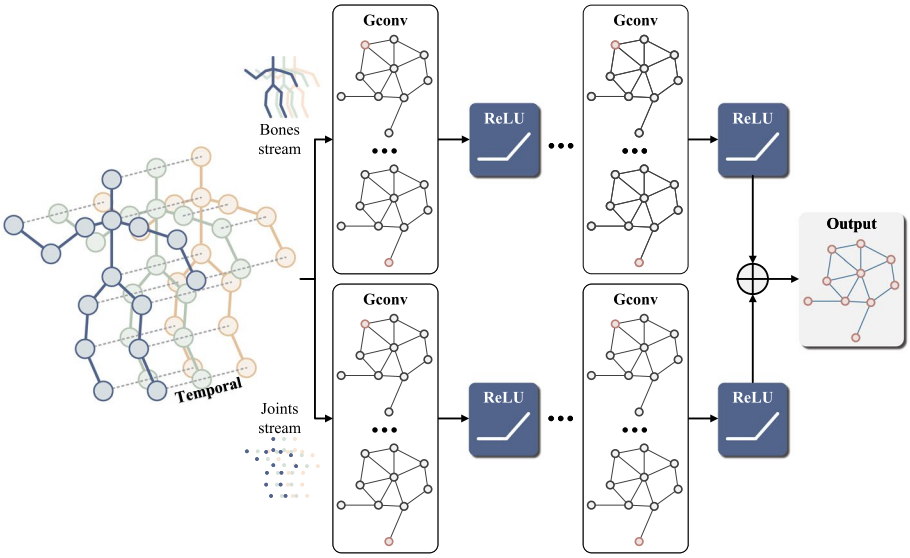

Fig. 8 Framework of multi-stream graph networks. Multi-stream graph networks attempt to model the complementary information of the body skeleton (joints and bones) for action recognition. The multi-stream architecture consists of separate graph convolution streams, which are subsequently concatenated and classified using the Softmax function

In the later stages of the network, the information from each stream is combined or fused to generate a unified representation for skeleton modeling. 2 s-AGCN (Shi et al. 2019) employs a dual-stream framework to simultaneously model first-order and second-order skeleton data, significantly improving the recognition accuracy. JOLO-GCN (Cai et al. 2021) integrates the human pose skeleton and joint-centric lightweight information within a dual-stream graph network, utilizing the joint flow patches (JFP) for the joint-aligned optical flow patch extraction to capture the subtle local motion around each joint. MS-AAGCN (Shi et al. 2020) models information about joints, skeletons, and their actions simultaneously within a multi-stream framework to capture more informative and discriminative features. RA-GCN (Song et al. 2021) employs a multi-stream GCN to explore distinguishing features distributed across all skeletal joints, where each stream learns features from currently inactive joints distinguished by activation maps obtained from previous streams. GCN-HCRF (Liu et al. 2021) utilizes a multi-stream framework that processes the relative coordinates of joints and bone orientations as two static feature streams, and the temporal displacement between two consecutive frames as a dynamic feature stream, preserving the structural information of the human skeleton throughout the classification stage. SEFN (Kong et al. 2021) is a multi-stream network composed of the spatial, temporal, and fusion streams that uses GCNs to extract the spatial features. The temporal stream employs the motion sequence calculation models to extract the temporal features, and the fusion stream applies a multi-perspective attention fusion module (MPAFM) to enhance feature extraction through an early fusion approach and additional fusion information.

3.5 Directed graph networks

Directed graph networks (DGNs) belong to the category of "Graph Structure-Based Frameworks." Unlike traditional graph networks that assume the influence between nodes is mutual and symmetric, the core of this category lies in constructing and leveraging a directed graph to model asymmetric relationships between nodes. As shown in Fig. 9, directed edges can represent physical causality, such as a shoulder movement causing an elbow movement, a hierarchical dependency, or the flow of kinetic energy. By using an asymmetric adjacency matrix where $A \neq A^T$, the model can learn these directional dependency patterns, providing a more refined tool for modeling the intrinsically hierarchical structure of the human skeleton.

Compared to the methods that only model individual bones, DGNs can capture the motion patterns of skeletons more effectively, providing superior motion representations for tasks such as skeleton-based human pose estimation and action recognition. Shi et al. (2019) introduced the DGNN network at CVPR 2019 as the first to incorporate DGNs into the skeletal modeling. This directed GNN was specifically designed to extract information about joints, bones, and their relationships while also modeling the spatial and temporal aspects of skeletons. PGCN (Bin et al. 2020) establishes the directed graphs between key joints and mapped them to structurally aware the key point representations, encoding both the skeletal structure and appearance information of specific key joints. DDGCN (Korban and Li 2020) captures the spatio-temporal dependencies, hierarchical structures, and sequential patterns in human behavior represented by skeletal graphs, contributing to improved modeling and recognition of human actions. ASE-GCN (Xiong et al. 2023) proposes the directed graph mapping (DGM) as a regularizer for skeleton structure information to enhance the graph structure representations by associating the directed graphs with human motion informa-

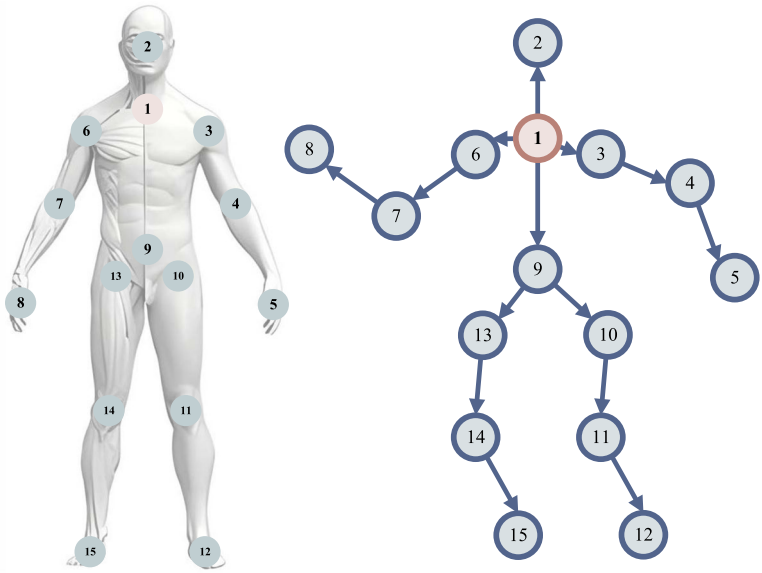

Fig. 9 Framework of directed graph networks. The red circles represent the root vertices. DGNs are designed for directed acyclic graphs (DAGs) and can model the tree structure of skeleton data, which aims to constrain the modeling to the local joints of the human body

tion using kinematic principles. Experiments have demonstrated that the skeleton modeling methods based on DGNs can achieve the state-of-the-art results on multiple datasets.

3.6 Dynamic graph networks

Dynamic graph networks fall under the category of "graph structure-based frameworks". These networks focus on dynamically adjusting graph topology over time to capture temporal variations in skeletal relationships. As shown in Fig. 10, unlike static methods that use a predefined graph of physical connections, dynamic graph networks allow the adjacency matrix itself to be a learnable, or even a time-variant (A_t), parameter. This enables the model to capture transient, functional, or context-dependent relationships that go beyond physical connections, such as the temporary strong correlation between two hands during a "clapping" motion. This significantly enhances the expressive power of the graph.

ST-GCN (Yan et al. 2018) proposed a coupling mechanism between spatial and temporal structure graphs for simultaneous learning of spatial and temporal features. DGNN (Shi et al. 2019) utilizes temporally varying edges along with "CONTEXT" to learn the correlation between sequential skeletons. These methods can more accurately and coherently analyze complex skeletal motion processes than static graph networks that model only single-frame information. SGSN (Li et al. 2021) introduces a dynamic multi-scale GNN (DMGNN) for predicting the human motion based on 3D skeletons, exhibiting the adaptability during training and dynamicity between network layers. LDRGCN (Cui et al. 2020) represents the skeleton as a novel dynamic graph that explicitly leverages the natural connectivity of joint pairs and implicitly learns the relationships between geometrically separated joints. DGCN (Qiu et al. 2020) employs the dynamic graph convolutional modules (DGCM) to model rich relationships in keypoint graphs and learn the structural relations from multi-level features, constructing dynamic graphs to accommodate significant variations in human poses. NAS-GCN (Peng et al. 2020) captures spatio-temporal correlations between the nodes and the multi-hop modules involving the high-order connections through

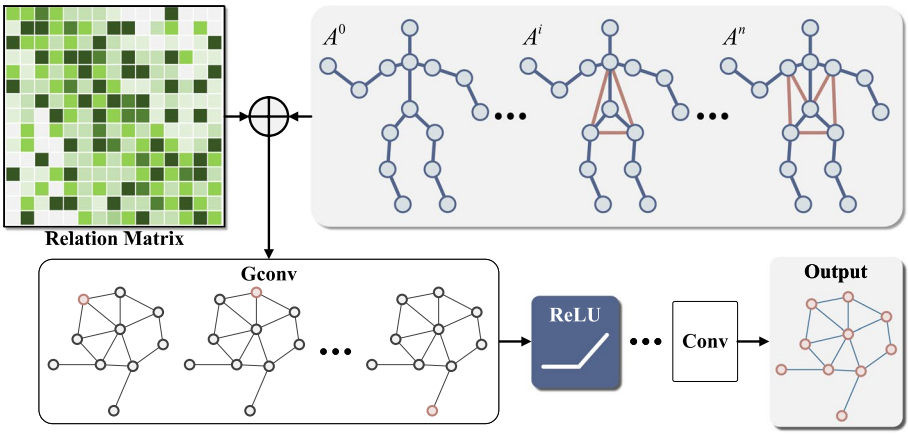

Fig. 10 Framework of dynamic graph networks. Constrained features A^i update aggregate pose representations with spatially shared correlations. The relation matrix obtains pose representations with specific spatial correlations, and the correlations are obtained through spatial correlation modeling. The output features were obtained through constrained graph networks using a correlation matrix

multiple dynamic graph modules. Skeletal-GNN (Zeng et al. 2021) proposes a time-aware dynamic graph construction method that learns the dynamic graphs evolving adaptively based on the input skeletons without relying on fixed predefined graphs.

In summary, graph networks adopt various structural forms, such as spatio-temporal graphs, directed graphs, and multi-scale graphs, in HSM. These structural forms enable the graph networks to exhibit several unique advantages.

(1) *Spatio-temporal information extraction*. Graph networks can capture spatio-temporal features between human joints and skeletons, including the positions and motion states of joints across different frames: (Kerola et al. 2017; Li and Leung 2017; Si et al. 2019; Li et al. 2019; Wu et al. 2019; Li et al. 2020; Korban and Li 2020; Zhang et al. 2020; Song et al. 2021; Huang et al. 2020; Mao et al. 2019; Liu et al. 2021; Kong et al. 2021; Miao et al. 2022; Setiawan et al. 2022; Liu et al. 2022, 2023; Zhu et al. 2023; Ma et al. 2022; Zhong et al. 2022; Song et al. 2023; Huang et al. 2023).
(2) *Learning complex non-Euclidean dependency relationships*. By learning the complex non-Euclidean dependency relationships between spatio-temporal skeletal joints, graph networks enable a comprehensive understanding of the dynamic evolution of the skeleton structure: (Yi and Lin 2016; Wang et al. 2016; Yang et al. 2021; Zhao et al. 2019; Shi et al. 2019; Liu et al. 2019; Cui et al. 2020; Bin et al. 2020; Peng et al. 2020; Li et al. 2020; Cai et al. 2019; Wu et al. 2021; Yu et al. 2023; Gao et al. 2023; Chi et al. 2022; Sofianos et al. 2021; Zhong and Cao 2023; Fu et al. 2023; Li et al. 2022; Doering et al. 2018; Chen et al. 2021).
(3) *Encoding hierarchical spatial structures*. By introducing multi-scale and multi-level node representations into the graph structure, graph networks effectively capture the hierarchical structure of human skeletons, allowing for a simultaneous focus on both the local details of the skeletal structure and overall motion. This comprehensive understanding of skeletal features is critical for the complex human action analysis and the skeletal modeling tasks: (Dang et al. 2021; Wen et al. 2019; Liu et al. 2020b, d; Zeng et al. 2020; Peng et al. 2020; Chen et al. 2021; Peng et al. 2021; Wu et al. 2022; Xiong et al. 2023; Wen et al. 2023; Zhang et al. 2023).
(4) *Learning long-range dependency relationships between human joints*. Graph networks aggregate node features at different distances to capture the global correlations between joints, enabling more accurate and reliable capture of the long-range dependencies between joints (Zou et al. 2020; Zhang et al. 2020; Liu et al. 2020e; Geng et al. 2022; Liu et al. 2023).
(5) *Hidden node inference and estimation*. Graph networks enable a comprehensive understanding of the human skeleton structure and learn the high-order relationships between joints to infer and estimate joint information that is not directly observable in data. This capability provides a robust performance for the skeleton modeling under conditions of incomplete or noisy data (Qiu et al. 2020; Ou et al. 2022; Yenduri et al. 2022; Li et al. 2019; Zhao et al. 2019; Zhou et al. 2021; Qin et al. 2022).

Owing to these advantages, graph networks have been applied to various HSM tasks, demonstrating unique strengths. In the human pose estimation task, graph networks learn the constraint relations between the skeletal joints, providing structural priors for the pose and enhancing the regression accuracy and robustness. In action recognition tasks, spatio-tem-

poral graph networks effectively model the temporal dynamics of the skeleton to identify complex actions. For human motion prediction, dynamic GATs can learn the relationships between bones at different times to enable the multi-step inference for predicting future skeletal motion trends. In pose estimation and tracking, graph networks offer overall skeletal structure constraints that can handle the occlusion situations and enhance the tracking stability. In summary, graph networks can provide the effective and unified modeling frameworks for a variety of skeletal-based human analysis tasks by learning complex skeletal structure constraints and dynamic information, offering the significant support for the skeletal representation.

3.7 Datasets and evaluation metrics

HSM is a core problem in computer vision and has been utilized to analyze and predict the human poses in images or videos. The construction of high-quality pose datasets is crucial for model training and evaluation. As shown in Table 2, datasets of human skeleton keypoint data are designed for human pose estimation tasks, focusing on identifying skeletal keypoint locations within individual image frames. These datasets typically provide spatial coordinates of skeletal keypoints (in 2D or 3D) for single-frame images, without incorporating temporal information or annotations for specific action categories. Commonly used datasets in this category include MSCOCO keypoints (Lin et al. 2014), MPII (Andriluka et al. 2014), and LSP (Johnson and Everingham 2010), which serve as benchmarks for evaluating static pose estimation methods. Datasets of human skeleton action data are tailored for action recognition tasks, emphasizing the temporal dynamics of human skeletal joints. These datasets capture sequences of skeletal keypoints over time (e.g., 3D joint trajectories) and are typically annotated with action labels. By leveraging the temporal evolution of joint positions, they enable the analysis and classification of human actions. Representative datasets include NTU RGB+D (Shahroudy et al. 2016), Kinetics Skeleton (Xia et al. 2012), and Human3.6M (Ionescu et al. 2013), which are widely adopted for benchmarking dynamic behavior recognition algorithms. The following are the introductions to several widely used and influential datasets in the field.

MSCOCO (Lin et al. 2014), MPII (Andriluka et al. 2014), and CrowdPose (Li et al. 2019) have extensively used image datasets for human pose estimation tasks. These datasets share the common characteristic of containing images with complex real-world backgrounds, where objects exhibit varying degrees of occlusion and background interference, to simulate practical scenarios for model applications.

NTU RGB+D (Shahroudy et al. 2016), NTU-RGBD 120, Kinetics-skeleton (Xia et al. 2012), MSR Action3D (Li et al. 2010), UCF101 (Soomro et al. 2012), and Northwestern-UCLA (Zhang et al. 2018) are video datasets widely used for human action recognition tasks. These datasets share common characteristics, including the presence of complex and diverse human actions in real-world settings, covering multiple categories such as daily activities and sports. Data sources include the internet, RGB/depth cameras, and other methods. The large volume of data and number of categories ensure both sample diversity and task difficulty. Some datasets provide multimodal information (RGB, depth, skeleton data, etc.) to entirely represent the human motion, whereas others capture actions using only RGB information. These real-world datasets serve as representative sources for the action-recognition tasks and are extensively used in model research.

Table 2 Comparison among different public datasets and metrics on human skeleton modeling

Model	Year	Institution	URL	Type	Number	Category	Metrics
Human skeleton keypoint data							
LSP (Johnson and Everingham 2010)	2010	University of Leeds	https://dbcollection.readthedocs.io/en/latest/datasets/leeds_sports_pose_extended.html	Image	2000	14	PCK, PCP
FLIC (Sapp and Taskar 2013)	2013	University of Pennsylvania	https://gas.graviti.cn/dataset/data-decorators/FLIC	Image	5003	10	PCK, PCP
MPII (Andriluka et al. 2014)	2014	Max Planck Institute for Informatics	http://human-pose.mpi-inf.mpg.de	Image	25,000	69	PCK
Human3.6M (Ionescu et al. 2013)	2014	Swiss Federal Institute of Technology	http://vision.imar.ro/human3.6m/description.php	Image	3.6M	24	MPJPE
CMU Panoptic (Joo et al. 2015)	2016	Carnegie Mellon University	http://domedb.perception.cs.cmu.edu/	Video	65	–	MPJPE
SYSU (Hu et al. 2015)	2016	Sun Yat-sen University	http://www.sysu-hcp.net/person/research/	Image	18,184	20	–
MSCOCO (Lin et al. 2014)	2017	Cornell University Microsoft Research Carnegie Mellon University	http://cocodataset.org/	Image	330K	17	AP, AR
MuCo-3DHP (Mehta et al. 2018)	2018	Max Planck Institute for Informatics	https://vcai.mpi-inf.mpg.de/projects/SingleShotMultiPerson/	Image	–	15	PCK, MPJPE
PoseTrack (Andriluka et al. 2018)	2018	Google Research	https://posetrack.net/	Video	514	15	AP, AR
OCHuman (Zhang et al. 2019)	2019	Tsinghua University	https://github.com/liruilong940607/OCHumanApi	Image	4731	17	AP, AR
CrowdPose (Li et al. 2019)	2019	Shanghai Jiao Tong University Tsinghua University	https://github.com/Jeff-sjtu/CrowdPose	Image	20,000	14	PCK
Human skeleton action data							
CMU Mocap (Li et al. 2018)	1999	Carnegie Mellon University	http://mocap.cs.cmu.edu/	Video	2605	8	MAE, MPJPE
HDM05 (Müller et al. 2007)	2007	Max Planck Institute for Informatics	http://resources.mpi-inf.mpg.de/HDM05/	Video	2337	130	–
HumanEva (Sigal et al. 2010)	2010	Carnegie Mellon University Massachusetts Institute of Technology	http://humaneva.is.tue.mpg.de	Video	7	6	MPJPE
MSR Action3D (Li et al. 2010)	2010	Microsoft Research	https://documents.uow.edu.au/~wanqing/#Datasets	Video	567	20	AP, AR, F1

Table 2 (continued)

Model	Year	Institution	URL	Type	Number	Category	Metrics
HMDB51 (Kuehne et al. 2011)	2011	Karlsruhe Institute of Technology assachusetts Institute of Technology Brown University	http://serre-lab.clps.brown.edu/resources/ HMDB/	Video	6849	51	–
UCF101 (Soomro et al. 2012)	2012	University of Central Florida	http://crcv.ucf.edu/data/UCF101	Video	13,320	101	Top-1, Top-5
Florence 3D Actions (Seidenari et al. 2013)	2012	University of Florence	https://www.micc.unifi.it/resources/datasets/florence-3d-actions-dataset/	Video	215	9	–
Human3.6M (Ionescu et al. 2013)	2014	Swiss Federal Institute of Technology	http://vision.imar.ro/human3.6m/description.php	Image	3.6M	17	MPJPE
UTD MHAD (Chen et al. 2015)	2015	University of Texas	https://personal.utdallas.edu/~kehtar/UTD-MHAD.html	Video	861	27	–
MPI-INF-3DHP (Mehta et al. 2017)	2016	Max Planck Institute for Informatics	http://gvv.mpi-inf.mpg.de/3dhp-dataset/	Video	896	8	PCK, MPJPE
Charades (Sigurdsson et al. 2016)	2016	Stanford University Meta/Facebook AI Reaserch	https://prior.allenai.org/projects/charades	Video	9848	157	AP, AR
NTU RGB+D 60 (Shahroudy et al. 2016)	2016	Nanyang Technological University	https://rose1.ntu.edu.sg/dataset/actionRecognition/	Image	56,880	60	X-Sub, X-View
Something-Something V1 (Goyal et al. 2017)	2017	University of Tabriz	https://github.com/open-mmlab/mmaction2/tree/master/tools/data/sthv1	Video	108,499	174	Top-1, Top-5
Something-something V2 (Goyal et al. 2017)	2017	University of Tabriz	https://github.com/open-mmlab/mmaction2/tree/master/tools/data/sthv2	Video	220,847	174	Top-1, Top-5
Kinetics-skeleton (Xia et al. 2012)	2018	–	https://github.com/yysijie/st-gcn	Image	260,232	400	Top-1, Top-5
3DPW (Von Marcard et al. 2018)	2018	Leibniz University Hannover	https://virtualhumans.mpi-inf.mpg.de/3DPW/	Video	60	18	MAE, MPJPE
Northwestern-UCLA (Zhang et al. 2018)	2018	Northwestern University University of California	https://wangjiangb.github.io/my_data.html	Video	1494	10	AP, AR, F1

Table 2 (continued)

Model	Year	Institution	URL	Type	Number	Category	Metrics
NTU RGB+D 120 (Liu et al. 2019)	2019	Nanyang Technological University	https://rose1.ntu.edu.sg/dataset/actionRecognition/	Image	114,480	120	X-Sub, X-Set
Action genome (Ji et al. 2020)	2019	Stanford University	https://www.actiongenome.org/	Image	234K	157	–

Human3.6M (Ionescu et al. 2013), CMU Mocap (Li et al. 2018), HumanEva-I/II (Sigal et al. 2010), and 3DPW (Von Marcard et al. 2018) are widely used action datasets for human motion prediction tasks. These datasets share the common characteristic of containing human motion data in real-world settings, covering a range of categories, such as daily activities, with both authenticity and diversity. These are often collected using multi-view setups and provide 2D or 3D human joint information, which is beneficial for motion predictions. Some datasets utilize the motion capture technology for high-precision data, whereas others use data from Internet videos. The large number of samples in these datasets meet the demands of model research.

Several metrics were used to evaluate the performance of the HSM models.

For pose estimation, the mean per joint position error (MPJPE) (Wang et al. 2021a) calculates the Euclidean distance between the estimated 3D joints and the ground truth positions, reflecting the accuracy of the 3D pose estimation. The percentage of correct parts (PCP) (Wang et al. 2021a) is determined when the distance between the predicted and ground truth joints was less than a fraction of the limb length, typically between 0.1 and 0.5. The percentage of correct keypoints (PCK) (Wang et al. 2021a) measures the accuracy of keypoint localization within a certain threshold, which is set to 50% of the head segment length of each test image. The intersection over union (IoU) (Wang et al. 2021b) identifies differences between the ground-truth annotations and predicted bounding boxes, with unnecessary predicted boxes removed based on a designated threshold (typically 0.5). The object keypoint similarity (OKS) (Wang et al. 2021b) is the average similarity of all object keypoints, calculated based on the subject's scale and the distance between the predicted points and ground truth points using scale and keypoint constants to equalize the weight of each keypoint. The average precision (AP) measures the accuracy of keypoint detection based on Precision, representing the average precision of all recall values between 0 and 1 across various IoU thresholds. The average recall (AR) evaluates the accuracy of keypoint detection based on the recall, representing the average recall of all recall values between 0 and 1 across various IoU thresholds.

Several evaluation metrics are commonly used for human-action recognition. Accuracy is the most common metric and is calculated as the proportion of correctly classified samples. F1 score (Li et al. 2010) represents the harmonic mean of the precision and recall, providing a single measure that balances the false positives and the false negatives. The precision gauges the proportion of the accurate positive identifications among all the positive predictions made by the model. X-View (cross-view) (Shahroudy et al. 2016) involves training the model with data from certain camera views and testing it with data from previously unseen camera views. X-Sub (cross-subject) (Shahroudy et al. 2016; Liu et al. 2019) partitions the dataset into the training and test sets, based on subject identities. X-Set (cross-

setup) (Liu et al. 2019) separates the dataset based on the variations in experimental setups, including differences in the environment, camera views, and other conditions. Top-1 and Top-5 metrics are straightforward indicators of performance, representing the percentage of instances in which the model's highest and top five probability outputs, respectively, include the correct label.

The human motion prediction extensively employs MPJPE and the mean angle error (MAE) (Lyu et al. 2022) to measure the joint coordinate errors and joint angle errors, respectively, to evaluate the accuracy of the prediction results. These metrics provide the quantitative and comprehensive comparisons of the model performance, driving the advancement of various tasks.

4 Application

4.1 Pose estimation and tracking

Significant progress has been made in the application of HSM in the field of pose estimation. Recently, inspired by the success of graph networks, a series of graph-network-based HSM and pose-estimation models have emerged. By learning the structure of human skeletons and the complex dependencies among joints using graph networks, these models can achieve accurate human pose estimation with promising applications in fields such as surveillance cameras and virtual reality. As shown in Fig. 11, the pose estimation methods based on graph network learning can entirely utilize the feature information in images or videos to enhance the model robustness, demonstrating better resistance to disturbances caused by factors such as perspective and occlusion.

4.1.1 2D pose

Different from typical convolutional networks only learning local features, graph networks offer a unique structured representation learning capability by modeling the non-Euclidean

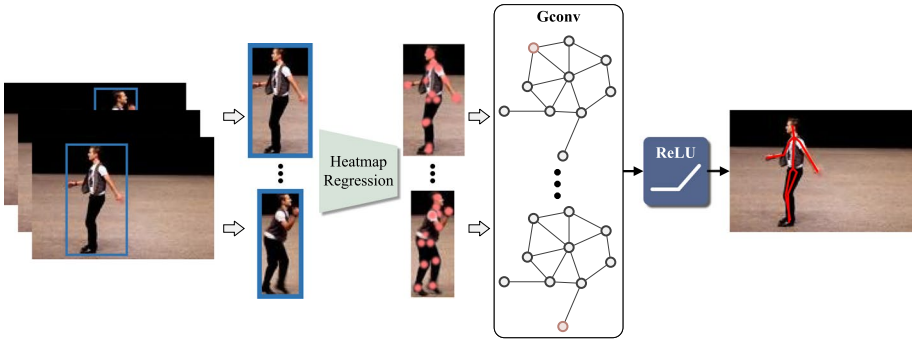

Fig. 11 Framework of pose estimation models based on graph networks. Pretrained object detectors are used to detect human bodies and provide human proposals. The human proposal regions are cropped from the original image to form individual images. Each cropped image underwent keypoint detection to output heatmaps, and the graph network encoded and connected the keypoints for single-person pose estimation

geometric spatial constraints between skeleton joints. As shown in Table 3, graph networks primarily serve two purposes: enhancing the modeling of skeletal structures and achieving joint multi-task learning with other tasks.

Occlusion poses a significant challenge for human pose estimation. Tian et al. (2021) have developed two simple yet effective modules, including the cascaded feature network (CFN) and the graph structure network (GSN), to strengthen the localization of visible and occluded joints. Graph-PCNN (Wang et al. 2020) employs a graph-network-based framework with a localization subnet and a graph-network-based pose refinement module to enhance the localization accuracy. The localization subnet extracts different visual features for each guidance point and explores the relationships between these points through a graph-network-based pose refinement module to achieve more accurate localization results. OPEC-Net (Qiu et al. 2020) introduces an image-guided progressive GCN module that comprehensively understands the image background and pose structure to estimate occluded joints from an inferential perspective. Fu et al. (2015) have proposed a model structure that encodes the interactions between human skeletal parts and objects, learns the occlusion consistency from data, and demonstrates the robustness to occlusion in 2D human pose estimation.

Table 3 Comparison with graph networks-based 2D pose estimation methods on MSCOCO (Lin et al. 2014) datasets in AP and AR

Model	Year	Category	Source	Backbone	Code	MSCOCO					
						AP	AP^{50}	AP^{75}	AP^{M}	AP^{L}	AR
DGCN (Qiu et al. 2020)	2020	Dynamic	AAAI	ResNet-50	–	67.4	88.0	74.4	63.6	73.0	73.2
Gragh-PCNN (Wang et al. 2020)	2020	Spatio-Temporal	arXiv	ResNet-50	–	76.8	92.6	84.3	73.3	82.7	81.6
HGG (Jin et al. 2020)	2020	Dynamic	arXiv	Hourglass	–	67.6	85.1	73.7	62.7	74.6	71.3
OPEC-Net (Qiu et al. 2020)	2020	Spatio-Temporal	arXiv	AlphaPose+	√	73.9	91.9	82.2	–	–	–
PGCN (Bin et al. 2020)	2020	Directed	PR	ResNet	–	71.1	–	–	–	–	77.0
MVG (Chen et al. 2020)	2020	Multi-Stream	ECCV	YOLO-v3	√	65.3	80.1	72.2	74.1	68.3	–
Tian et al. (2021)	2021	Spatio-Temporal	PR	ResNet-50	–	74.4	–	–	–	–	–

Several improved solutions based on graph networks have achieved notable results in 2D pose estimation (Bin et al. 2020). CPF-PGNN (Zhang et al. 2019) introduces two conceptually simple yet computationally efficient modules, cascade prediction fusion (CPF) and pose GNN (PGNN), to leverage the contextual information for human pose estimation. DGCN (Qiu et al. 2020) considers all relationships between keypoints and constructs a dynamic graph to accommodate the significant variations in human poses. HGG (Jin et al. 2020) redefines the human part grouping as the graph clustering problem and introduces a novel differentiable layer graph grouping method. HGG can be easily integrated into the mainstream bottom-up methods and employs multi-layer GNN models for clustering the keypoints. In summary, graph networks provide a unique structural representation learning mechanism for 2D pose estimation, enhancing both the constraint modeling of the skeleton and joint learning capability with related tasks.

4.1.2 3D pose

Graph networks have been widely applied to 3D human pose estimation tasks by fully utilizing the non-Euclidean geometric relationships between skeletal joints. As shown in Table 4, they can contribute to two major aspects: including the establishment of the mapping between image features and 3D skeletons and the provision of prior spatial structural knowledge among skeletal joints.

GTRS (Zheng et al. 2021) proposed a pose analysis module utilizing the graph transformers to leverage the structured and implicit joint correlations, and introduces a mesh regression module for the human mesh reconstruction. Modulated-GCN (Zou and Tang 2021) addresses the limitations of previous GCN methods by introducing the weight modulation and the affinity modulation to enhance the feature transformations among different nodes. Skeletal-GNN (Zeng et al. 2021) presents a novel GNN learning solution that includes a skip-aware hierarchical channel squeeze fusion layer and a time-aware dynamic graph construction procedure. SemGCN (Zhao et al. 2019) captures semantic information such as local and global node relationships that are not explicitly represented in the graph. Wu et al. (2021) proposed a multi-view matching graph (MMG) module for learning the cross-view correlations and recovering the coarse human proposals, while the center refinement graph (CRG) module can refine the results through flexible point-based predictions. For pose estimation, the pose regression graph (PRG) module learns the structural relationships between multi-view geometric graphs and human joints. Cai et al. (2019) integrated spatio-temporal relationships by adjusting the graph topology based on the input poses for adaptive graph topology. These methods can overcome the limitations of traditional graph networks and enhance the pose estimation performance.

Graph structure modeling is a key component of the graph-network-based 3D pose estimation. Choi et al. (2020) uses 2D human poses as the input to provide basic skeletal joint information and employs GraphCNN to utilize the mesh topology in a coarse manner. SRNet (Zeng et al. 2020) divides the skeleton into the local regions and simulates rare and occluded poses using splitting and recombination methods. Zhao et al. (2021) introduces the GraAttention module to address the gaps in graph structure information. These methods can offer different approaches to graph structure modeling and present potential pathways for improving the pose estimation performance. LGPF-GCN (Liu et al. 2020d) learns the global pose features in GCNs by adding a global node connected to all skeletal joints and

Table 4 Comparison with graph networks-based 3D pose estimation methods on Human 3.6M (Ionescu et al. 2013) and MPI-INF-3DHP (Mehta et al. 2017) datasets. MPJPE results (in mm) follow the standard protocol without rigid alignment. Results are cited from original papers; omissions occur if a method was not evaluated on a specific benchmark in its publication

Model	Year	Category	Source	Backbone	Code	Human 3.6M		MPI-INF-3DHP				
						#1 MPJPE	#2 MPJPE	GS	noGS	Outdoor	PCK	AUC
SemGCN (Zhao et al. 2019)	2019	Spatio-Temporal	CVPR	ResNet-50	√	42.5	–	–	–	–	–	–
Cai et al. (2019)	2019	Dynamic	ICCV	–	–	48.8	39.0	–	–	–	–	–
LCN (Ci et al. 2022)	2019	Spatio-Temporal	ICCV	–	√	32.5	27.1	74.8	70.8	77.3	74.0	36.7
High-order GCNs (Zou et al. 2020)	2020	Multi-Scale	BMVC	ResNet-101-FPN	–	55.6	43.7	–	–	–	–	–
Liu et al. (2020b)	2020	Attention	arXiv	ResNet-101-FPN	√	45.7	35.9	–	–	–	–	–
LGPF-GCN (Liu et al. 2020d)	2020	Multi-Scale	ACCV	–	–	54.9	42.8	79.0	79.3	79.8	79.3	45.9
Choi et al. (2020)	2020	Spatio-Temporal	arXiv	HRNet ResNet-50	√	64.9	48.7	–	–	–	–	–
SRNet (Zeng et al. 2020)	2020	Multi-Scale	ECCV	FCN	√	33.9	–	–	–	80.3	77.6	43.8
Wu et al. (2021)	2021	Multi-Stream	ICCV	–	–	15.84	–	–	–	–	–	–
Modulated-GCN (Zou and Tang 2021)	2021	Spatio-Temporal	ICCV	–	√	49.4	39.1	86.4	86.0	85.7	86.1	53.7
Skeletal-GNN (Zeng et al. 2021)	2021	Dynamic	ICCV	–	√	47.9	30.4	–	–	84.6	–	–
Azizi et al. (2022)	2022	Spatio-Temporal	ECCV	–	–	40.0	–	79.2	77.3	83.1	80.0	–
Zhao et al. (2021)	2022	Attention	CVPR	–	√	35.2	–	80.1	77.9	74.1	79.0	43.8
GTRS (Zheng et al. 2021)	2022	Attention	arXiv	ResNet-50	√	50.3	30.4	–	–	–	–	–
Zhang et al. (2023)	2023	Attention	IJCV	–	√	49.1	34.9	88.7	88.5	81.49	86.9	55.0
GCAT-Net (Qiao et al. 2025a)	2025	Attention	IJCV	SemGCN+CCA+Transformer	–	41.7	31.7	–	–	–	–	–

proposes a heterogeneous multitask learning framework to learn the local and global features. High-order GCNs (Zou et al. 2020) learn the long-range dependency relationships between human joints, aggregate the node features at different distances, and capture the remote dependencies between joints.

Owing to their inherent weight-sharing scheme, GCNs have faced significant limitations in 3D pose estimation. LCN (Ci et al. 2022) addresses these limitations by using a GCN to estimate 3D poses from 2D poses and assigns dedicated filters to different nodes rather than sharing filters. Another major limitation of GCNs is their inability to encode all transformations between the nodes explicitly. To address this issue, MobiusGCN (Azizi et al. 2022) introduces the Möbius transformation to directly and explicitly encode the transformations between joints. The innovation of this method lies in its ability to explicitly represent all transformations between joints.

A detailed analysis of Tables 3 and 4 reveals several key trends in graph-based pose estimation. Early methods like SemGCN (Zhao et al. 2019) demonstrated the foundational viability of GCNs by modeling semantic relationships between joints, achieving competitive accuracy. However, recent top-performing methods such as PHGANet (Zhang et al. 2023) and Modulated-GCN (Zou and Tang 2021) signify a clear shift toward more sophisticated architectures. PHGANet's success on the challenging MPI-INF-3DHP dataset, particularly its state-of-the-art 86.9 PCK, highlights the benefit of its multi-hop graph attention mechanism, which captures complex, long-range dependencies that simpler models miss. Similarly, the effectiveness of Skeletal-GNN (Zeng et al. 2021) and its time-aware dynamic graph construction underscores the importance of adaptive topologies that can change based on the input pose, a crucial feature for handling diverse and difficult poses in the wild. This trend indicates that the field is moving beyond fixed graph structures toward dynamic and attention-aware models to achieve higher accuracy and robustness.

The results in Table 4 reveal that SOTA performance in pose estimation is achieved through two distinct pathways: innovating on the input data representation or innovating on the graph network architecture itself. The remarkably low error of Wu et al. (2021) on Human3.6M illustrates the first pathway. Its success is not solely from its Graph Network, but primarily from its use of multi-view video data. The proposed Multi-View Matching Graph effectively fuses information from multiple camera angles, fundamentally reducing the depth ambiguity that challenges single-view methods. This highlights that enriching the input data can yield significant performance gains. In contrast, Zhang et al. (2023) exemplifies the second pathway, achieving state-of-the-art results on the challenging MPI-INF-3DHP benchmark through pure Graph Network architectural innovation. Its hierarchical graph attention module is specifically designed to capture complex, multi-hop dependencies between joints, demonstrating that a more sophisticated understanding of skeletal topology directly translates to higher accuracy, especially on in-the-wild data.

A dominant trend is the synergistic fusion of GNNs and Transformers to harness their complementary strengths. GNNs excel at encoding local, relational inductive biases inherent in the human skeleton, while transformers are powerful at capturing long-range, global dependencies across all joints. A prime example of this paradigm is GCAT-Net (Qiao et al. 2025a), a novel framework that integrates a semantic GCN (SemGCN) for local topology modeling, criss-cross attention (CCA) for capturing long-range spatial dependencies, and a transformer encoder to model the temporal evolution of the pose sequence. This hybrid architecture allows the model to build a comprehensive understanding of the pose, from

fine-grained local joint relationships to global body configuration and temporal dynamics, achieving a state-of-the-art MPJPE of 41.7 mm on the Human3.6M benchmark (Qiao et al. 2025a). Furthermore, the scope of HSM is expanding from analyzing isolated human motion to understanding interactions within a broader context. The GeoVis-GNN model (Qiao et al. 2025b) pioneers this direction for multi-person human-object interaction (HOI) recognition. It employs a bottom-up approach where a dual-attention mechanism first fuses geometric features and visual features to create robust representations for each entity. Subsequently, an "interdependent entity graph" models the higher-level interactions. This represents a significant step towards heterogeneous graph modeling, where nodes of different types are reasoned over jointly, enabling a more semantic, context-aware understanding of human actions.

4.2 Action recognition

Graph networks offer unique advantages in human action recognition tasks thanks to their powerful relational modeling capabilities. Compared with the traditional CNN-based methods that focus only on local information, graph networks can capture the structural information of all actions by modeling the relationships between human keypoints. Specifically, as shown in Fig. 12, graph networks construct a graph, where the nodes represent the keypoint features and edges represent the relationships between these keypoints. This approach enables the graph networks to learn the overall structure of human actions and to effectively represent the relationships between different actions. In this subsection, we reviewed the studies on graph network-based action recognition, as detailed in Table 5.

ST-GCN (Yan et al. 2018) introduces a spatio-temporal GCN designed to improve the message aggregation by eliminating redundant embeddings of nodes, frames, or elements across different network layers, which can address the issue of the over-smoothing in graphs for action recognition and provide efficient and accurate graph embeddings. Additionally, several research efforts have extended ST-GCN by utilizing GCNs to learn the spatial and temporal features for the action recognition at multiple scales (Chen et al. 2021), applying attention mechanisms (Yu et al. 2023; Heidari and Iosifidis 2021) to capture the spatio-temporal dependencies between joints (Ghosh et al. 2020; Peng et al. 2020; Wu et al. 2019). In contrast, the introduction of a series of GCN+LSTM networks for action recognition from the skeletal data has yielded empirical recognition results. Through utilizing GCN, these networks can extract the structure-aware feature representations from skeletal data and employ the LSTM networks to capture the spatio-temporal dynamics of skeletons (Si et al. 2019; Zhao et al. 2019; Huang et al. 2020; Li et al. 2020; Si et al. 2018). Additionally,

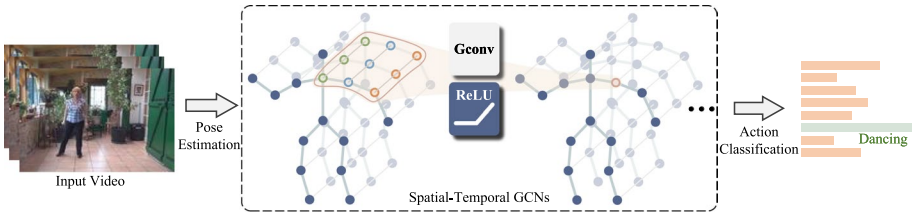

Fig. 12 Framework of action recognition models based on graph networks. Multi-layer spatio-temporal GCNs were used to progressively generate higher-level feature maps on the graph. Finally, a standard softmax classifier was employed to classify the corresponding action categories

Table 5 Comparison with graph networks-based action recognition methods on NTU RGB+D 60 (Shahroudy et al. 2016), NTU RGB+D 120 (Liu et al. 2019) and Kinetics (Xia et al. 2012) datasets. For NTU datasets, X-Sub and X-View/X-Set represent cross-subject and cross-view/setup protocols, respectively. All accuracies are reported as Top-1 classification accuracy (%)

Model	Year	Category	Source	Code	NTU RGB+D 60		NTU-RGB+D 120		Kinetics	
					X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
PB-GCN (Thakkar and Narayanan 2018)	2018	Spatio-Temporal	BMVC	√	87.3	93.6	–	–	–	–
Si et al. (2018)	2018	Spatio-Temporal	ECCV	–	84.8	92.4	–	–	–	–
ST-GCN (Yan et al. 2018)	2018	Spatio-Temporal	AAAI	√	81.5	88.3	–	–	30.7	52.8
DPRL+GCNN (Tang et al. 2018)	2018	Spatio-Temporal	CVPR	–	83.5	89.8	–	–	–	–
A^2GNN (Li et al. 2018)	2018	Spatio-Temporal	IEEE TIP	–	72.74	82.8	–	–	–	–
motif-GCN (Wen et al. 2019)	2019	Multi-Scale	AAAI	–	84.2	90.2	–	–	–	–
Shi et al. (2019)	2019	Multi-Stream	CVPR	√	88.5	95.1	–	–	36.1	58.7
Si et al. (2019)	2019	Attention	CVPR	–	89.2	95	–	–	–	–
AS-GCN (Li et al. 2019)	2019	Dynamic	CVPR	√	86.8	94.2	–	–	34.8	56.5
Zhao et al. (2019)	2019	Spatio-Temporal	ICCV	–	81.8	89	–	–	–	–
Shi et al. (2019)	2019	Directed	CVPR	√	89.9	69.1	–	–	36.9	59.6
STGR (Li et al. 2019)	2019	Dynamic	AAAI	–	86.9	92.3	–	–	33.6	56.1
Wu et al. (2019)	2019	Spatio-Temporal	ICCV	–	89.58	95.74	–	–	37.35	60.38
Si-GCN (Liu et al. 2019)	2019	Spatio-Temporal	IJCNN	–	84.15	89.05	–	–	–	–
Liu et al. (2020c)	2020	Multi-Scale	CVPR	√	91.5	96.2	86.9	88.4	38	60.9
CA-GCN (Zhang et al. 2020)	2020	Spatio-Temporal	CVPR	–	83.5	91.4	–	–	34.1	56.6
DDGCN (Korban and Li 2020)	2020	Directed	ECCV	√	91.05	97.14	–	–	38.12	60.79
GECNN (Zhang et al. 2020)	2020	Spatio-Temporal	IEEE TNNLS		85.4	91.1	–	–	33.4	56.2
Shi et al. (2020)	2020	Multi-Stream	IEEE TIP	–	90	96.2	–	–	37.88	61.13

Table 5 (continued)

Model	Year	Category	Source	Code	NTU RGB+D 60		NTU-RGB+D 120		Kinetics	
					X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
MV-IGNet (Wang et al. 2020)	2020	Multi-Stream	IEEE TPAMI	√	89.2	96.3	83.9	85.6	–	–
PL-GCN (Huang et al. 2020)	2020	Spatio-Temporal	AAAI	–	89.2	95	–	–	–	–
Shift-GCN (Cheng et al. 2020a)	2020	Spatio-Temporal	CVPR	√	90.7	96.5	85.9	87.6	–	–
TL-GCN (Zhu et al. 2020)	2020	Dynamic	PRL	–	89.2	95.4	–	–	36.2	59
RA-GCN (Song et al. 2021)	2020	Multi-Stream	IEEE TCSVT	√	87.3	93.6	81.1	82.7	–	–
Huang et al. (2020)	2020	Spatio-Temporal	WACV	–	84.71	91.74	–	–	–	–
Miki et al. (2020)	2020	Spatio-Temporal	WACV	–	79.9	89.8	–	–	–	–
Cheng et al. (2020b)	2020	Spatio-Temporal	ECCV	–	90.8	96.6	86.5	88.1	–	–
NAS-GCN (Peng et al. 2020)	2020	Dynamic	AAAI	√	89.4	95.7	–	–	37.1	60.1
Li et al. (2020)	2020	Spatio-Temporal	WACV	–	86.9	92.5	–	–	–	–
TE-GCN (Li et al. 2020)	2020	Dynamic	arXiv	–	90.8	96.2	84.4	85.9	–	–
VE-GCN (Liu et al. 2020e)	2020	Multi-Scale	ISCAS	–	90.1	95.2	84.5	85.2	–	–
SGP+JCA (Chen et al. 2020)	2020	Spatio-Temporal	PR	–	86.9	93.8	–	–	–	–
CTR-GCN (Chen et al. 2021)	2021	Dynamic	ICCV	√	92.4	96.8	88.9	90.6	–	–
Chen et al. (2021)	2021	Multi-Scale	AAAI	√	91.5	96.6	87.5	88.8	38.1	60.8
Tripool (Peng et al. 2021)	2021	Spatio-Temporal	PR	–	88	95.3	80.1	82.8	34.1	56.2
Heidari and Iosifidis (2021)	2021	Spatio-Temporal	IJCNN	–	87.8	95.1	–	–	–	–
GCN-HCRF (Liu et al. 2021)	2021	Multi-Stream	IEEE TMM	–	90	95.5	–	–	–	–
SEFN (Kong et al. 2021)	2021	Multi-Stream	IEEE TCSVT	–	90.7	96.4	86.2	87.8	39.3	62.1

Table 5 (continued)

Model	Year	Category	Source	Code	NTU RGB+D 60		NTU-RGB+D 120		Kinetics	
					X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
JOLO-GCN (Cai et al. 2021)	2021	Multi-Stream	WACV	–	93.8	98.1	87.6	89.7	38.3	62.3
Hyper-GNN (Hao et al. 2021)	2021	Multi-Scale	IEEE TIP	–	89.5	95.7	–	–	37.1	60
Sybio-GNN (Li et al. 2022)	2022	Spatio-Temporal	IEEE TPAMI	–	90.1	95.4	–	–	37.2	58.1
FGCN (Yang et al. 2022)	2022	Spatio-Temporal	IEEE TIP	–	90.2	96.3	85.4	87.4	–	–
OR^{2}G (Ou et al. 2022)	2022	Spatio-Temporal	CVPR	–	91	96.5	86.3	87.8	–	–
CD-GCN (Miao et al. 2022)	2022	Spatio-Temporal	IEEE TCSVT	√	90.9	96.5	86.3	87.8	–	–
Qin et al. (2022)	2022	Multi-Scale	IEEE TNNLS	√	88.7	94.5	83.2	83.7	–	–
Setiawan et al. (2022)	2022	Spatio-Temporal	EXPERT SYST APPL	–	92.8	96.9	–	–	38.1	62.9
Graph2Net (Wu et al. 2022)	2022	Multi-Scale	IEEE TCSVT	–	90.1	96	86	87.6	–	–
Liu et al. (2022)	2022	Attention	KNOWL-BASED SYST	–	90.4	96.1	86.1	88	38.1	61
InfoGCN (Chi et al. 2022)	2022	Attention	CVPR	√	93	97.1	89.8	91.2	–	–
Liu et al. (2023)	2023	Attention	IEEE TVCG	–	90.7	96.1	86.3	87.8	37.8	60.9
ASE-GCN (Xiong et al. 2023)	2023	Directed	IEEE TCSVT	–	89.4	96.2	–	–	36.9	59.7
Zhu et al. (2023)	2023	Multi-Scale	IEEE TIP	–	91.85	96.18	88.6	90	38.9	62.2
Huang et al. (2023)	2023	Multi-Stream	IEEE TCSVT	–	90.8	96.4	87	88.4	38.7	62.3
Wu et al. (2023)	2023	Multi-Scale	PR	–	91.1	96.5	86.5	88.2	36.1	58.9
DGCN (Jia et al. 2023)	2023	Dynamic	PRL	–	89.2	96.3	83.9	85.6	–	–
Yu et al. (2023)	2023	Multi-Stream	IEEE TPAMI	–	96	98.8	92.9	94.4	–	–
SAMotif-GCN (Wen et al. 2023)	2023	Multi-Scale	IEEE TPAMI	√	90.5	96.1	87.1	87.7	37.8	60.6
HetGCN (Gao et al. 2023)	2023	Spatio-Temporal	IEEE TNNLS	–	93.1	97.2	88.9	90.3	38.4	61.2

Table 5 (continued)

Model	Year	Category	Source	Code	NTU RGB+D 60		NTU-RGB+D 120		Kinetics	
					X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
Song et al. (2023)	2023	Attention	IEEE TPAMI	√	92.1	96.1	88.7	88.9	–	–

attention mechanisms (Song et al. 2023; Chi et al. 2022), multi-scale graph networks (Wu et al. 2022; Wen et al. 2019, 2023; Gao et al. 2023), and multi-stream graph networks (Shi et al. 2019, 2020; Song et al. 2021; Liu et al. 2021; Kong et al. 2021) have been adopted to enhance the model's action representation capability and recognition accuracy.

Owing to the irregular nature of the variations in human motion processes, a series of adaptive learning graph network methods have been proposed to enhance the flexibility and versatility of graph-network-based approaches. GIN (Tang et al. 2020) focuses on "self-learning" weight matrices to represent the pairwise relationships between different nodes in the graph. CTRGCN (Chen et al. 2021) dynamically learns various topologies and efficiently aggregates the joint features across different channels, easing the stringent constraints on GCNs and improving the action representation capabilities. This self-learning process enables the exploration of potential graph topologies beyond the manually crafted designs, enhancing the performance of skeleton-based action recognition (Zhu et al. 2020).

Some studies have focused on extracting high-order connectivity relationships from skeletons (Peng et al. 2020) to capture the action-specific latent dependencies and the higher-order interactions. Qin et al. (2022) extend the joint neighbors and combine high-order skeletons to create new high-order adjacency matrices. A^2GNN (Li et al. 2018) constructs each irregular skeleton as an undirected attribute graph and performs the local spectral graph filtering to extract the advanced semantic representations of actions. STGR (Li et al. 2019) introduced the spatio-temporal graph routing for the skeleton-based action recognition, adapting to the high-order connectivity relationships between joints. GHRM (Zhou et al. 2021) integrates the information from all basic relations when modeling each basic relation to leverage high-order relations for long-term action recognition. PL-GCN (Huang et al. 2020) learns the high-level relations between the skeletal parts and emphasizes the important parts of the actions. Hyper-GNN (Hao et al. 2021) represents the human skeleton as a hypergraph and uses the hyperedge structures to extract both local and global structural information, capturing the spatio-temporal information and the high-order dependencies to mitigate the impact of noise from irrelevant joints.

For human skeletons, the nodes can exhibit the long-range dependency relationships, which are crucial for capturing the semantic clues for key actions. Several action recognition methods have been proposed to capture these relationships effectively. CA-GCN (Zhang et al. 2020) employs the asymmetric correlation metrics and the higher-order representations to consolidate information from all vertices, computing contextual information. VE-GCN (Liu et al. 2020e) performs the GCN operations on specified vertices, incorporating additional connections between joints to capture distant dependencies. STPG-Net (Geng et al. 2022) adopts a GCN for the long-range relationship inference on spatio-temporal pyramid graphs, which are designed to enhance action representations for distinguishing video actions.

Some studies have focused on the interactions between different parts of the human skeleton by capturing the internal patterns through the subgraph construction. SPGK (Wang et al. 2016) represents the video motion elements as an undirected, fully labeled graph, where the decomposed subgraphs correspond to the comparisons between video sub-actions. PB-GCN (Thakkar and Narayanan 2018) uses the subgraphs for each part of the skeleton to compose the overall skeleton graph, performs the GCN operations separately on each subgraph, and aggregates the results for the inter-subgraph information fusion, which enables the learning of part importance and inter-part relationships. Si-GCN (Liu et al. 2019) formulates an inter-graph to model the relationships between these partial subgraphs, using the spectral GCNs to capture the internal patterns of each human part and consider the interactions among different parts.

Some other works have proposed novel research approaches, including separable parameter GCN (SPGconv) (Wang et al. 2020), shift graph operation and lightweight pointwise convolution (Cheng et al. 2020a), similarity graph (Miki et al. 2020), graph pooling (Chen et al. 2020; Peng et al. 2021), Symbiosis Model (Li et al. 2022), Feedback GCN (Yang et al. 2022), and other improved methods of GCN. These methods have considered the graph topology of the human skeleton and global graph context from various perspectives to capture motion patterns between skeletal joints, enhancing the ability of GCNs to infer node correlations and recognize human motion patterns.

Table 5 provides a comprehensive overview of the rapid advancements in skeleton-based action recognition. A clear evolutionary path is visible. The pioneering ST-GCN (Yan et al. 2018) established a strong baseline, demonstrating the power of jointly modeling spatial and temporal dimensions. Subsequent innovations fall into patterns identified by our taxonomy. Methods like 2 s-AGCN (Shi et al. 2019) and MS-AAGCN (Shi et al. 2020) introduced multi-stream and attention-enhanced adaptive graphs, significantly boosting performance by focusing on more informative joints and motion features. The exceptional results of InfoGCN (Chi et al. 2022) and CTR-GCN (Chen et al. 2021), which consistently lead the NTU leaderboards, highlight a recent trend: the move toward learning the graph topology itself in a task-driven manner. Instead of relying on a fixed physical skeleton, these models learn latent dependencies, allowing for more flexible and powerful representations.

The results in Table 5 clearly show that the highest echelon of action recognition performance is now dominated by models that synergistically combine graph networks with other data modalities or learning paradigms. The standout performance of MMNet (Yu et al. 2023), which achieves a remarkable 96.0% accuracy on NTU RGB+D 60, is a sophisticated multi-modal fusion framework. MMNet uses the attention maps generated by an ST-GCN on skeleton data to guide a separate network in focusing on the most salient regions within the RGB video frames. This allows the model to ground the abstract skeletal dynamics in rich, contextual visual evidence, leading to a significant performance leap. This case demonstrates that the frontier has moved beyond intra-skeleton modeling to GNN-guided cross-modal analysis. The outstanding performance of JOLO-GCN (Cai et al. 2021) warrants a closer look as it exemplifies this multi-faceted success. Its significant performance gain is not derived solely from a better GCN architecture, but from a synergistic combination of architecture and feature engineering. Its key innovation, lies in one of these streams, which processes "Joint Flow Patches." This feature leverages optical flow to capture subtle, pixel-level local motion dynamics around each joint. This information serves as a powerful supplement to pure skeleton coordinates, allowing the model to perceive fine-grained move-

ments that are difficult to describe with joint positions alone. While InfoGCN (Chi et al. 2022) achieves top results through innovations within the GCN paradigm itself, such as its mutual information-based learning objective, the exceptional success of MMNet underscores a key trend: the most significant breakthroughs often arise from using graph networks not as isolated processors of skeleton data, but as intelligent guides for richer data sources.

Recent research continues to push the boundaries by addressing the inherent limitations of both GCNs and Transformers. GCNs are often constrained by a predefined skeletal topology, struggling to model interactions between non-adjacent but functionally related joints, while Transformers, with their global attention, can sometimes accumulate noise by modeling irrelevant dependencies. To address this, the dynamic graph attention network (DGAN) has been proposed (Li et al. 2025a). DGAN dynamically constructs an action-specific adjacency matrix by employing a masked attention mechanism. This allows the network to flexibly model both local physical connections and crucial long-range correlations, effectively integrating the strengths of graph attention and dynamic graph paradigms. Concurrently, to mitigate the reliance on large labeled datasets, self-supervised learning has become a key research direction. The Multi-Skeleton Contrastive Learning (MS-CLR) framework (Zhang et al. 2025b) exemplifies this trend. It learns robust and generalizable representations by enforcing consistency between different skeleton formats extracted from the same video sequence. This encourages the model to learn structural invariances and diverse anatomical cues, improving performance in downstream action recognition tasks.

4.3 Motion Prediction

The human motion prediction is a prominent research area in the fields of computer vision and artificial intelligence. As a powerful method for data modeling and representation, graph networks depict human skeletal joints as a graph structure, learning joint features and spatio-temporal relationships to offer distinctive advantages in human motion prediction. As illustrated in Fig. 13, graph networks can aggregate the neighbor information to learn the joint characteristics and the global context, predicting future motions for each joint via classifiers. Temporal information can be incorporated into the spatio-temporal GCNs. The main advantage of graph networks in human motion prediction is their ability to handle the complex non-Euclidean structured data and capture the spatio-temporal relationships between human joints. Compared to the traditional sequential models, graph networks not only model the connections between joints, but also adaptively learn the importance and interaction between the nodes, enhancing the accuracy and robustness of motion prediction. In addition, graph networks can learn both the global and local features to effectively manage the irregular action sequences and adapt to various types and lengths of motion sequences.

Most research strategies can model the dynamics of human skeletons and the spatial dependencies of joints using graph networks that automatically learn the graph connectivity. These strategies employ the nodes in a graph for the spatio-temporal evolution to effectively describe the motion data and enhance the subsequent motion prediction results (Mao et al. 2019; Sofianos et al. 2021; Ma et al. 2022; Li et al. 2022; Xu et al. 2022). Several proposals utilize multiple scales of spatio-temporal graphs for feature extraction, transitioning from the fine-grained to the coarse-grained scales and back to the fine-grained scales (Dang et al. 2021; Li et al. 2021). DMGNN (Li et al. 2020) employs the multiscale graph networks to

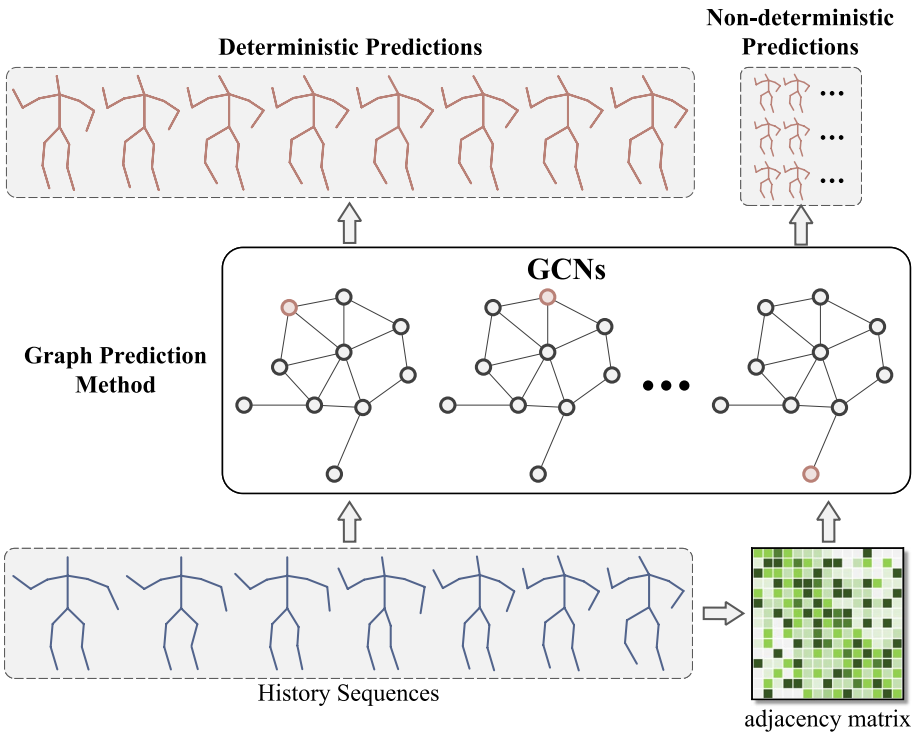

Fig. 13 Framework of motion prediction models based on graph networks. The task of human motion prediction is divided into three stages: input of human pose sequences, graph prediction model, and prediction results. A human pose sequence is a parameterized observation of the physical or mathematical transformations of a sequence. The graph prediction model aims to extract and process features to facilitate the prediction. For a given historical sequence, multiple future frames of prediction results are generated. Deterministic prediction assumes that the future motion sequence is singular and uniquely determined by the given input sequence. The model outputs a single future motion trajectory, without accounting for potential variability or uncertainty in future actions. Non-deterministic predictions acknowledge the inherent diversity and uncertainty in future motions. The model predicts multiple plausible future motion trajectories, capturing the range of possible developments

dynamically extract and fuse the cross-scale features using an encoder-decoder framework where the encoder consists of a series of mgCUs for learning the motion features, and the decoder applies the graph-network-based gated recurrent units to generate future poses. In LDRGCN (Cui et al. 2020), the skeletal poses are represented as the novel dynamic graph that exploits the natural connectivity of joint pairs to indirectly learn the links between geometrically distinct joints, with the graph networks and adversarial learning used to enhance the accuracy and visualization.

Other improvements based on graph networks have also been reported. GA-MIN (Zhong and Cao 2023) proposes a novel geometric algebra multiview interaction network that enhances the motion prediction performance by combining GCN with other techniques. Because using GCNs for feature filtering within a finite graph spectrum may lead to the information loss across the entire bandwidth, SPGSN Li et al. (2022) suggests employing an adaptive graph-scattering method. This method decomposes the skeletal features into richer spectral bands using multiple trainable bandpass-graph filters. Additionally, MTGCN

(Liu and Yin 2020) introduces a lightweight framework based on a multi-granular trajectory GCN for the person-independent human motion prediction, representing the human motion as the multi-branch trajectories, including joint and sub-joint trajectories.

The results in Table 6 for motion prediction highlight the critical challenge of handling long-term dependencies. A clear pattern emerges: while most models perform reasonably well on short-term predictions, their error rates increase significantly for long-term horizons (1000 ms). This indicates the difficulty of preventing error accumulation over time. Notably, methods that explicitly model graph structure beyond simple kinematics, such as LDRGCN (Cui et al. 2020) which learns long-range dependencies, and STSGCN (Sofianos et al. 2021) with its spatio-temporal separable graph, show a distinct advantage in long-term accuracy on the Human3.6M benchmark. STARS (Xu et al. 2022) achieves an impressive 44.5 MPJPE at 320 ms, suggesting its trajectory-space representation is effective for mid-term prediction. The performance of GA-MIN (Zhong and Cao 2023) on the 3DPW dataset further underscores the importance of integrating geometric algebra principles, which may offer more robust rotational representations. Overall, the most successful methods are those that innovate on the graph's definition to capture complex, non-local, and long-range spatio-temporal relationships, which is essential for mitigating error propagation in this generative task.

After analyzing the results in Table 6, the models that best mitigate long-term issue do so through significant architectural innovations. STSGCN (Sofianos et al. 2021) achieves strong long-term performance on Human3.6M by introducing a spatio-temporal separable GCN. This architectural design explicitly disentangles spatial and temporal correlations, preventing the feature contamination that can accelerate error accumulation in deeper models. This shows a direct link between a specific Graph Network architecture improvement and better long-term stability. Another top performer, GA-MIN (Zhong and Cao 2023), showcases a different path to success by incorporating principles from outside standard graph network design. Its use of geometric algebra provides a more robust mathematical framework for handling rotations and joint relationships, which standard vector-based approaches may struggle with. The success of these methods indicates that overcoming the long-term prediction challenge requires either deep innovations within the graph network architecture to better manage information flow, like in STSGCN, or the integration of powerful external mathematical tools to create more robust representations, as seen in GA-MIN.

The recent literature introduces novel architectures designed to further address these challenges. The spatio-temporal multi-subgraph GCN (STMS-GCN) (Guo et al. 2025) proposes a sophisticated framework that decouples spatial and temporal modeling into two parallel branches. Inspired by the Mixture-of-Experts paradigm, it utilizes multiple trainable subgraphs to capture a richer variety of motion patterns. Crucially, it enforces a spatio-temporal information consistency constraint between the two branches, facilitating cross-domain knowledge transfer and enhancing the model's ability to capture complex dependencies (Guo et al. 2025). Another significant advancement is the application of diffusion models to this generative task. The spatio-temporal graph motion diffusion (STGMD) model (Liu et al. 2023) employs a hybrid architecture where a GNN-based U-Net (STG-UNet) captures fine-grained local joint interactions, while an auxiliary Transformer aggregates global context and handles conditioning on textual descriptions. The non-autoregressive nature of the diffusion process can inherently mitigate the error accumulation problem that plagues

Table 6 Comparison with graph networks-based motion prediction methods on Human 3.6M (Ionescu et al. 2013), CMU Mocap (Li et al. 2018) and 3DPW (Von Marcard et al. 2018) datasets. MPJPE and MAE are reported in millimeters and degrees, respectively, for short-term and long-term prediction horizons. Lower values indicate better performance

Model	Year	Category	Source	Human 3.6M (MPJPE)						CMU Mocap (MPJPE)					3DPW (MPJPE)							
				80	160	320	400	560	1000	80	160	320	400	1000	100	200	400	500	600	800	900	1000
Traj-GCN (Mao et al. 2019)	2019	Spatio-Temporal	ICCV	12.1	25.0	51.0	61.3	50.4	71.0	11.5	20.4	37.8	46.8	62.8	16.28	35.62	67.46	80.19	90.36	106.79	113.93	117.84
Li et al. (2020)	2020	Dynamic	CVPR	16.95	33.62	65.9	79.65	93.57	127.62	14.07	24.44	45.90	55.45	104.33	17.8	37.11	70.38	83.02	94.12	109.67	117.25	123.93
LDRGCN (Cui et al. 2020)	2020	Dynamic	CVPR	10.69	22.55	45.13	55.79	–	97.83	9.36	17.59	31.63	43.18	82.9	–	33.9	57.4	–	84.6	95.2	–	109.1
MTGCN (Liu and Yin 2020)	2020	Multi-Scale	arXiv	10.1	22.7	47.7	58.6	78.2	111.9	8.5	15	32.7	42.8	94.7	–	–	–	–	–	–	–	–
Dang et al. (2021)	2021	Multi-Scale	ICCV	12.11	25.56	51.64	62.93	62.89	86	8.72	15.83	30.57	38.1	78.63	15.7	33.48	65.02	77.59	93.81	108.15	114.88	116.31
SGSN (Li et al. 2021)	2021	Dynamic	ICCV	10.6	22.9	47	56.9	67.3	100.8	9.1	16	32.4	41.3	86	14.9	32.3	62.2	–	91.5	103.3	107	110
Sofianos et al. (2021)	2021	Spatio-Temporal	ICCV	10.1	17.1	33.1	38.3	50.8	75.6	–	–	–	–	–	18.32	37.79	67.51	77.34	92.75	106.65	113.14	112.22
Li et al. (2022)	2022	Spatio-Temporal	ECCV	10.44	22.33	47.07	58.26	77.4	109.64	8.3	14.8	28.64	36.96	77.82	15.39	32.91	64.54	76.23	91.62	103.98	109.41	111.05
Ma et al. (2022)	2022	Spatio-Temporal	CVPR	10.3	22.7	47.4	58.5	76.9	110.3	7.6	14.3	29	36.6	80.1	–	29.3	58.3	–	79.8	94.4	–	104.1
Xu et al. (2022)	2022	Spatio-Temporal	ECCV	9.7	21.2	44.5	55.5	75.1	108.4	–	–	–	–	–	–	–	–	–	–	–	–	–
Fu et al. (2022)	2022	Spatio-Temporal	arXiv	10.38	23.34	48.77	59.84	77.81	111.02	7.33	13.9	28.37	35.91	79.95	–	–	–	–	–	–	–	–
GAGCN (Zhong et al. 2022)	2022	Attention	CVPR	10.1	16.9	32.5	38.5	50	72.9	–	–	–	–	–	–	–	23.6	–	–	–	–	39.9
GA-MIN (Zhong and Cao 2023)	2023	Spatio-Temporal	PR	9.4	19.9	42.4	52.2	68	102.2	8.7	15.4	31.9	40.1	84.5	–	27.2	57.1	–	71.2	82.1	–	90.6
Model	Year	Category	Source	Human 3.6M (MAE)						CMU Mocap (MAE)					3DPW (MAE)							
				80	160	320	400	560	1000	80	160	320	400	1000	100	200	400	500	600	800	900	1000
Traj-GCN (Mao et al. 2019)	2019	Spatio-Temporal	ICCV	0.27	0.51	0.83	0.95	0.9	1.27	0.25	0.39	0.68	0.79	1.33	–	0.64	0.95	–	1.12	1.22	–	1.27
Li et al. (2020)	2020	Dynamic	CVPR	0.27	0.52	0.83	0.95	0.89	1.21	0.23	0.35	0.69	0.82	1.32	–	–	–	–	–	–	–	–
LDRGCN (Cui et al. 2020)	2020	Dynamic	CVPR	0.26	0.45	0.71	0.86	–	1.2	0.23	0.295	0.58	0.72	1.23	–	0.57	0.72	–	1.07	1.18	–	1.25
Li et al. (2021)	2021	Multi-Scale	IEEE TIP	0.26	0.5	0.79	0.92	1.15	1.55	0.21	0.33	0.62	0.72	1.27	–	0.61	0.91	–	1.13	1.21	–	1.24
SGSN (Li et al. 2021)	2021	Dynamic	ICCV	0.25	0.48	0.79	0.91	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Li et al. (2022)	2022	Attention	IEEE Trans Cybern	0.2	0.41	0.59	0.69	0.83	1.11	0.24	0.4	0.7	0.82	1.2	–	–	–	–	–	–	–	–
GAGCN (Zhong et al. 2022)	2022	Attention	CVPR	0.24	0.38	0.54	0.65	0.74	1.02	–	–	–	–	–	–	–	–	–	–	–	–	–
GA-MIN (Zhong and Cao 2023)	2023	Spatio-Temporal	PR	0.23	0.45	0.79	0.89	1.08	1.49	0.19	0.31	0.58	0.71	1.27	–	0.62	0.89	–	1.10	1.18	–	1.21

traditional sequential models, offering a promising direction for more stable and realistic long-term motion prediction.

5 Discussion

Graph networks have achieved considerable success in the modeling human skeletal structures and their applications. They have been adopted for the pose estimation and action recognition because of their ability to effectively encapsulate the spatial information of the human skeleton (Simonyan and Zisserman 2014). Furthermore, their intrinsic ability to concurrently model the spatio-temporal skeletal structures over video sequences (Yan et al. 2018), has led to the integration of temporal information into GCNs through the inter-frame nodal connections for the spatio-temporal correlation modeling. The simultaneous occurrence of the spatio-temporal skeletal information has made GCNs a superior method compared to CNNs, RNNs, or their combination. The observations have indicated that recent research trends can pivot towards graph networks. Although the proposed methods have demonstrated the efficacy, several challenges persist and require further exploration.

5.1 Issues and challenges

While graph networks have pushed the boundaries of HSM, several fundamental challenges remain. In this section, we discuss these key limitations and propose potential mitigation strategies and future research directions for each.

The shallow structure of graph networks The traditional deep learning models based on CNNs typically achieve refined information features by stacking multiple convolutional layers, reducing the network parameters, and increasing the nonlinear transformations to enhance recognition performance (Li et al. 2017). However, in graph networks, experimental results have indicated a decline in performance when the number of convolutional layers exceeds two (Kipf and Welling 2016). This can be because an excessively deep stack of layers leads to the over-smoothing of each node within the graph, erasing distinctive features, and consequently hindering the recognition performance. To combat over-smoothing and enable deeper graph networks, researchers can draw inspiration from the success of deep CNNs. Incorporating residual or skip connections allows feature information from earlier layers to bypass intermediate layers, preserving the original node identity. Furthermore, employing attention mechanisms can help deeper layers to selectively focus on the most salient neighborhood information rather than uniformly averaging it. Advanced graph normalization techniques specifically designed to maintain feature variance across layers also present a promising avenue for building more powerful, deeper GCNs for HSM.

The diversity of human poses and appearances The fundamental challenge in developing accurate human skeletal models lies in the diversity of human skeletons (Moeslund and Granum 2001). Despite this variation, the human skeleton requires a high degree of freedom, necessitating the data-driven methods for HSM to possess sophisticated representational capabilities. Moreover, image degradations such as motion blur and out-of-focus issues present in video-based data (Luo et al. 2018) impede the performance of HSM methods. The adaptive capacity of graph topology is limited. During human motion, the graph topology changes with shifting actions, affecting the relationships between the nodes and

the selection of key nodes. When new connections are introduced or existing associations dissolve, the graph networks exhibit the limited adaptability. Although some adaptive studies have been conducted (Shi et al. 2019; Li et al. 2019), they are rule-based and cannot generate the on-the-fly dynamics. The most direct solution lies in the development of more sophisticated dynamic graph networks, as discussed in our taxonomy. These methods allow the graph's adjacency matrix to be learned or adapted on-the-fly based on the input pose, capturing transient dependencies. A complementary approach is the use of graph attention networks, which can be viewed as a "soft" dynamic mechanism. By learning to re-weight edge importance for each specific action, attention allows the model to dynamically adjust information flow on a fixed topology. Additionally, leveraging unsupervised pre-training on massive, unlabeled video datasets could enable models to learn more robust and generalizable motion priors, making them inherently less sensitive to pose variations.

Occlusion Current methods for HSM have exhibited the commendable performances on many public datasets. However, a well-acknowledged issue is the significant performance degradation caused by occlusions and the highly deformable nature of the human skeleton (Andriluka et al. 2014; Fang et al. 2017; Cheng et al. 2020). The self-occlusion, object occlusion, and mutual occlusion between multiple objects can prompt the modeling truncation, particularly in crowded scenes (Li et al. 2019), leading to a sharp decline in model performance. Additionally, highly deformable human skeletons can create the ambiguities in small-scale human instances or certain human poses (Li et al. 2019). Consequently, there is a need for an accurate HSM method to address the challenges posed by various occlusions and individual appearances, estimate the depth of monocular images.

The reason graph networks offer a fundamental improvement in occlusion resilience lies in their core operational differences from CNNs and RNNs. Methods based on CNNs operate directly on pixel grids and are highly dependent on local visual evidence. When a body part is occluded by an object, its corresponding pixel information is completely replaced, and the CNN loses all direct evidence for identifying that part because it has no inherent understanding of the body's global structural connectivity. On the other hand, methods based on RNNs typically process joint coordinates as a flattened sequence over time. When a joint's coordinates are missing or inaccurate due to occlusion, the input sequence itself is corrupted. Although an RNN can use its temporal memory to make an inference, it lacks an explicit model of the body's physical constraints and cannot understand that an occluded wrist's position is strongly constrained by the visible elbow and shoulder.

In contrast, the fundamental advantage of Graph Network is that its operational domain is a graph structure that explicitly encodes biomechanical constraints. The core mechanism of Graph Networks, message passing, allows the model to reason about a node (joint) even when its own feature information is missing. Specifically, when a wrist node is occluded, its connected and visible neighbors, such as the elbow and shoulder, can still pass "messages" containing their own positional and state information to it. The Graph Network, having learned the structural priors of the skeleton during training, can then perform effective relational reasoning based on these messages to infer a plausible position for the occluded joint. This ability to leverage global topological structure to compensate for missing local information is the fundamental reason for the innate occlusion resilience of Graph Networks compared to CNNs and RNNs.

5.2 Comparative analysis and practitioner's guide

While the previous sections have categorized and detailed the various graph network methods in HSM, a crucial question for any developer or researcher is how to choose among them. This section provides a head-to-head, critical comparison of the six categories in our taxonomy and offers a practice-oriented guide for method selection. We summarize the key characteristics of each category in Table 7.

As summarized in Table 7, a clear trade-off exists between performance, cost, and applicability for each category. Spatio-temporal networks are the cornerstone of the field, providing an efficient and powerful starting point for most tasks. Their primary limitation, a static graph structure, motivated the development of other categories. Graph attention networks and multi-scale networks can be seen as "modular" improvements upon a fixed topology, achieving a good balance between computational overhead and performance gain through more sophisticated feature aggregation.

Directed graph networks and dynamic graph networks represent more fundamental, "structural" improvements, attempting to make the graph itself more intelligent. Among these, dynamic graph networks are theoretically the most expressive and are best suited for scenarios with complex human-human or human-object interaction, where the effective connectivity between joints is constantly changing. This ultimate flexibility, however, comes with the highest computational cost and design complexity. Multi-stream networks offer a direct path to state-of-the-art performance, operating on the principle that more information is better. By processing and fusing different representations in parallel, such as joint positions and bone vectors, they achieve excellent results, almost always at the cost of multiplied computational resources.

Practitioner's takeaway Based on this analysis, a practitioner should not seek a single "best" model, but rather make a strategic decision by weighing the core requirements and resource constraints of their specific application.

For scenarios that demand a balance of performance and high efficiency, such as real-time applications like online pose tracking or large-scale video analysis, starting with a mature and lightweight spatio-temporal network is a wise and pragmatic choice. These models are generally straightforward to implement and deploy, and they serve as a powerful performance baseline for rapid project validation.

To achieve targeted accuracy improvements on top of this baseline, modular enhancements are the next logical step. If the application requires the precise capture of subtle or key-part motions, as found in medical rehabilitation analysis or human-computer interaction with fine-grained gestures, incorporating a graph attention mechanism is an effective upgrade path. It allows the model to adaptively focus on the most informative body parts, and its attention maps can sometimes offer a degree of interpretability. For tasks that demand an understanding of complex, full-body coordinated movements, such as dancing, sports analytics, or complex daily activities, Multi-Scale Networks hold an advantage due to their ability to simultaneously capture a hierarchy of dynamic features from local joints to entire limbs and the overall body.

When maximum accuracy is the paramount goal, and computational and development resources are ample, higher-complexity frameworks should be considered. If the application has access to multi-source or multi-modal data (e.g., bone vectors in addition to joint coordinates, or even the RGB stream), then multi-stream networks offer the most direct path to

Table 7 A comparative analysis of the six categories of graph network-based HSM methods

Category	Primary goal	Strengths and use cases	Limitations	Computational efficiency
Spatio-temporal graph	Model spatio-temporal dynamics of skeleton sequences	Strong performance baseline, suitable for general action recognition	Fixed topology is suboptimal for unconventional actions	High
Graph attention	Adaptive weighting of key nodes and edges	Captures subtle motions, enhances focus on critical body parts	Attention weights can be hard to interpret, adds computation	Medium to High
Multi-scale graph	Capture hierarchical features from joints to the whole body	Strong understanding of complex, hierarchical actions	Multi-scale design can introduce redundancy, complex to design	Medium
Multi-stream graph	Fuse information from different sources	Often achieves SOTA accuracy due to complementary information	Significant increase in parameters and computation, high training cost	Low
Directed graph	Model causal or hierarchical dependencies between joints	Expresses skeletal hierarchy, aligns with biomechanics	Predefined directionality may not suit all actions	High
Dynamic graph	Allow graph topology to adapt to the action itself	Most expressive for actions with drastic topological changes	Can be unstable to train, highest computational cost, complex design	Low

state-of-the-art performance by leveraging complementary information. Finally, if the core research challenge lies in navigating unconventional and drastic topological changes, such as multi-person grappling, complex tool manipulation, or deep human-environment interaction, then exploring dynamic graph networks is the most forward-looking and promising direction. Practitioners should be aware, however, that this ultimate flexibility often comes with the highest training difficulty and computational expense, positioning it more as a frontier for research exploration than a readily deployable industrial solution in most cases.

5.3 Future directions

Several relevant studies have been proposed that have produced notable results in addressing the aforementioned challenges. It is foreseeable that numerous future works will further explore graph networks to enhance the HSM applications.

Exploring novel node features and types of graphs Current graph convolutional networks are generally shallow structures, which are difficult to effectively capture deep-level node relationships and global graph structural information, limiting their performance on complex graph data. Therefore, mining new node features and establishing diverse types of graphs can be crucial in future studies. The flexibility of graph networks can be fully exploited to establish more types of graphs and analyze nonstructural data. For the HSM tasks, additional visual features, such as interactive features, scene graphs, and RGB characteristics, can be introduced along with the existing coordinates and appearance features at each node of the skeleton graph. Integrating the contextual information (e.g., image scenes, objects, and interactions) with the skeletal data in graph models has achieved considerable success in certain applications such as action recognition (Yan et al. 2018; Si et al. 2019; Dong et al. 2020). Combining these elements within the framework of graph networks and merging the interactions between human skeletons and scenes facilitates the enhanced graph-network-based skeletal modeling.

Unsupervised learning Fully supervised methods currently dominate the field of HSM because of their superior performance, which results from the rich postural annotations in large-scale datasets. However, unlabeled images and videos makes it impractical to provide thorough annotations for these datasets. A potential avenue for future research is the exploration of unsupervised or weakly supervised methods for the graph-network-based skeleton modeling. For instance, Qiu et al. (2020) devised GCC, a self-supervised graph representation learning framework designed to capture the universal network topological properties of multiple networks. Verma and Zhang (2019) proposed DUGNN, a deep universal graph embedding network that uses the graph kernels to learn the task-agnostic graph embeddings on different datasets. Shu et al. (2023) introduced a multi-granularity anchor contrastive representation learning approach, which learns the multi-granularity representations covering the local, contextual, and global views by performing the cross-granularity and intra-granularity contrastive prediction tasks on the learnable and structurally linked skeletons. Unsupervised learning capable of automatically learning human skeletons from limitless amounts of data has emerged as an important direction to consider.

Multimodality Occlusion and self-occlusion problems significantly limit the performance of the model in complex scenarios. Especially in multi-person scenarios or large changes in perspective, the model may not be able to accurately locate key points or predict skeletal movement. Future research can develop a multimodal learning framework to enhance the robustness of the model. In particular, the topological modeling ability of the graph network is used to effectively integrate skeletal information from different modes, so as to maintain high accuracy in the occlusion situation. At the same time, exploring long-range dependency modeling based on time-sequence relationships also helps to solve the problem of dynamic occlusion. Significant strides have been made through innovative strategies, such as attention mechanisms, multi-stream fusion, and dynamic configurations. Many models (Shi et al. 2020; Liu et al. 2020c; Cheng et al. 2020b) use fusion methods to integrate several input sequences into multi-stream architectures, combining the softmax scores of streams to achieve an aggregated score for action label prediction. Precise video-content comprehension can be substantially enhanced by leveraging the multimodality. Shi et al. (2020) demonstrate that each modality (skeleton, audio, text) brings distinct advantages, suggesting that more effective fusion strategies can elevate performance levels. Fan et al. (2020) suggested the effective inclusion of video subtitles as a source of contextual informa-

tion alongside RGB video streams. However, the challenges of multimodal fusion are non-trivial, primarily stemming from three key areas: feature alignment, modality imbalance, and the selection of fusion strategy. Feature alignment addresses the difficulty of mapping heterogeneous data–such as the graph-structured skeleton data and grid-like RGB data–into a common, meaningful representation space. Modality imbalance occurs when one modality dominates the learning process, potentially overshadowing the contributions from other, sparser modalities like skeleton data. Regarding fusion strategies, they are typically categorized as early fusion (combining raw data), late fusion (combining decision scores), and hybrid fusion (combining features at intermediate layers), with each presenting its own trade-offs between information richness and architectural complexity. To elicit more distinctive and rich features, current multi-stream fusion frameworks tend towards complexity and high parameterization. Cai et al. (2021) proposed the application of joint flow patch (JFP) sequences to encapsulate subtle local motion details, while AdaMML (Panda et al. 2021) dynamically adapts the mode selection for each video segment.

Model of artificial intelligence generated content (AIGC) for HSM Estimating human skeletal structures from natural scenes is crucial for visual perception. HSM plays a pivotal role in advancing fields such as artificial intelligence in AIGC and human-computer interaction, which enables deeper insights into human behavior in real-world environments. This burgeoning field empowers AIGC to synthesize highly realistic human motion, facilitating the applications ranging from immersive entertainment with expressive virtual characters to the ergonomic analysis for productivity enhancements. Integrating human posture information into large-scale language models (LLMs) enhances the semantic understanding, enabling them to recognize actions, intentions, and emotions. This advancement paves the way for more intuitive and impactful human-computer interaction. Xiang et al. (2023) introduced GAP for skeleton-based action recognition, which utilize a pretrained LLM to automatically generate the textual descriptions of action body part movements. RFID-ACCLDM (Wang and Mao 2024) generates the high-quality human activity data, serving as a viable substitute for real data in training human activity recognition models and achieving superior performance. However, significant challenges should be addressed to realize these advancements. Developing the data-efficient algorithms capable of generalizing across diverse modalities and environments remains critical. As research progresses in HSM within AIGC and LLM, balancing technological advancement with ethical considerations is essential to fully harness its transformative potential.

Heterogeneous graph modeling for contextual understanding Most current graph networks model the human skeleton as a homogeneous graph, where all nodes, being the joints, and edges, being the bones, are of a single type. This paradigm excels at capturing the internal kinematics of the human body but inherently treats the body as an isolated system. Human behavior, however, is deeply embedded in its surrounding context. To break this limitation, a significant trend is the evolution toward heterogeneous graph modeling. A heterogeneous graph allows the network to contain multiple types of nodes and edges. For instance, nodes can represent not only joints but also body parts, objects in the scene, or even abstract action concepts. Edges can then represent not only physical connections but also human-object interactions, semantic relationships between joints, or part-whole hierarchies.

The shift to heterogeneous graphs is significant because it enables graph networks to move beyond pure geometric analysis and begin performing true context-aware reasoning.

By explicitly integrating scene information into the graph structure, models can potentially handle occlusion more robustly or disambiguate actions by understanding human-object interactions. HetGCN (Gao et al. 2023) dynamically analyzes pairwise interactions and heterogeneous context aggregation between each node and its spatio-temporal neighbors to capture rich semantic representations from skeleton-based graphs. While research on explicitly constructing and applying large-scale heterogeneous graphs is still nascent, this direction represents a key future breakthrough for the HSM field: moving skeleton analysis out of a vacuum and into the rich, dynamic context of the real world.

Robust HSM with principled uncertainty modeling As previously discussed, occlusion and noise introduce significant data uncertainty in HSM. Current graph network models typically learn to handle this uncertainty in an end-to-end, implicit manner, for instance by directly regressing what they determine to be the most likely joint position. This approach, however, lacks formal theoretical guidance, and the reliability of its output can be questionable under high uncertainty. A more promising direction involves adopting principled uncertainty management frameworks from other mathematical fields.

Three-way decision theory offers such a powerful framework. It posits that when faced with uncertain information, a decision should not be a simple binary choice of acceptance or rejection, but should include a third option of delay, thereby partitioning elements into a positive region, a negative region, and a boundary region. In the context of HSM, this means a future model could process an occluded joint not by forcing a specific coordinate output, but by classifying its state into one of three categories: accepted as accurately located, rejected as noise or an error, or placed in a boundary state for deferment, requiring fusion with other modalities or evidence from subsequent frames for a final decision. This deferred decision mechanism can effectively prevent the model from making high-risk, erroneous judgments when information is insufficient. Recent advances in uncertainty research, such as 3WD-FGBRS (Yang et al. 2025a) and 3WC-SGBs (Yang et al. 2025b), offer sophisticated mathematical tools to precisely define and manage these uncertain boundary regions. Integrating these formal uncertainty modeling techniques into future Graph Network architectures is a promising interdisciplinary research direction that could lead to more robust and reliable HSM systems.

Standardized benchmarking and reproducibility studies. As our survey demonstrates, the field is advancing rapidly with a multitude of proposed methods. However, results are often reported on different datasets with varying evaluation protocols, making direct and fair comparison challenging. A large-scale study that re-implements key models under a single, unified framework to provide a truly objective comparison–as astutely suggested during our review process–would be a significant service to the community. Such an effort would validate reported results, fill gaps in the literature, and establish a more reliable benchmark for future work. We therefore identify this as a critical direction for the field's continued maturation.

6 Conclusion

This study provided a comprehensive overview of the graph-network-based methods for HSM. It systematically covered the fundamental concepts of HSM and reviewed six prominent frameworks of the graph networks used in the skeletal modeling. This discussion

included three key applications: pose estimation, action recognition, and motion prediction. Finally, we identified the research trends, addressed the challenging issues, and suggested the future research directions. To aid readers in implementing the state-of-the-art methods, the open-source codes from the notable graph-network-based HSM studies are referenced in Tables 3, 4, 6, facilitating the practical execution of HSM tasks. This synthesis served as both an introductory material for newcomers to the graph-network-based HSM and the reference materials for advanced researchers.

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Declarations

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